- **Parity Bit Positions:** Parity bits are placed at positions that are powers of 2 (1, 2, 4, 8, ...).
- **Parity Check:** Each parity bit checks specific data bit positions. For example, parity bit at position 2^k checks all positions whose binary representation has the k-th bit set.

Common Pitfalls and Problem-Solving Techniques:

- **Pitfall:** Incorrectly assigning parity bit positions or calculating parity values.
- **Technique:** Systematically determine parity bit positions, then calculate parity values based on the data bits they cover. For error detection/correction, calculate syndrome bits.

IP Addressing

IP addressing (IPv4 and IPv6) is a fundamental concept in the Network Layer, providing unique logical addresses for devices on a network. IPv4 uses 32-bit addresses, while IPv6 uses 128-bit addresses.

Important Formulas and Results (IPv4):

- **Number of Hosts:** For a given subnet mask, if h bits are available for host IDs, then the number of usable hosts is $2^h - 2$ (subtracting network and broadcast addresses).
- **Number of Subnets:** If s bits are borrowed from the host portion for subnetting, the number of subnets created is 2^s.
- **CIDR (Classless Inter-Domain Routing):** Uses a prefix length (e.g., /24) to specify the network portion of an IP address, replacing class-based addressing.

Key Properties and Identities:

- **Network ID:** All host bits are 0.
- **Broadcast ID:** All host bits are 1.
- **Private IP Ranges:** Reserved for private networks (e.g., 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16).

Common Pitfalls and Problem-Solving Techniques:

- **Pitfall:** Incorrectly calculating subnet masks, network IDs, or broadcast IDs, especially with CIDR notation.
- **Technique:** Convert IP addresses and subnet masks to binary for clear visualization. Practice subnetting problems extensively.

IP Packet

An IP packet (or datagram) is the fundamental unit of data transfer in the Internet Protocol. It consists of an IP header and a data payload.

Key Concepts (IPv4 Header Fields):

- **Version (4 bits):** Indicates IPv4 or IPv6.
- **Header Length (IHL, 4 bits):** Length of the IP header in 32-bit words. Minimum 5 (20 bytes), maximum 15 (60 bytes).
- **Total Length (16 bits):** Total length of the IP datagram (header + data) in bytes. Max 65535 bytes.
- **Identification (16 bits):** Used for reassembling fragmented datagrams.
- **Flags (3 bits):** Includes DF (Don't Fragment) and MF (More Fragments).
- **Fragment Offset (13 bits):** Position of the fragment in the original datagram (in 8-byte units).
- **Time To Live (TTL, 8 bits):** Decremented by each router; packet is discarded if TTL reaches 0. Prevents infinite loops.
- **Protocol (8 bits):** Indicates the next-level protocol (e.g., 6 for TCP, 17 for UDP, 1 for ICMP).
- **Header Checksum (16 bits):** Used for error detection only on the header. Recalculated at each hop.
- **Source IP Address (32 bits):** Sender's IP address.
- **Destination IP Address (32 bits):** Receiver's IP address.

Common Pitfalls and Problem-Solving Techniques:

- **Pitfall:** Confusing header length (in 32-bit words) with total length (in bytes).

- **Technique:** Memorize the key fields and their sizes. Practice calculating header checksums.

ICMP (Internet Control Message Protocol)

ICMP is a Network Layer protocol used by network devices, like routers, to send error messages and operational information indicating, for example, that a requested service is not available or that a host or router could not be reached.

Key Concepts:

- **Error Reporting:** Reports errors such as "Destination Unreachable," "Time Exceeded," "Parameter Problem."
- **Query Messages:** Used for diagnostic purposes (e.g., Echo Request/Reply for ping, Timestamp Request/Reply).
- **Traceroute:** Uses ICMP Time Exceeded messages to map the path to a destination.

Common Pitfalls and Problem-Solving Techniques:

- **Pitfall:** Confusing ICMP's role with routing protocols or transport layer protocols. ICMP is for network diagnostics and error reporting.
- **Technique:** Understand the common ICMP message types and their uses (e.g., ping uses type 8/0, traceroute uses type 11).

LAN Technologies

LAN (Local Area Network) technologies define the standards and methods for connecting devices within a limited geographical area. Ethernet is the dominant LAN technology.

Key Concepts:

- **Ethernet (IEEE 802.3):** Dominant wired LAN technology, uses CSMA/CD (for shared media) or full-duplex switching. Various speeds (10 Mbps, 100 Mbps, 1 Gbps, 10 Gbps, etc.).
- **Token Ring (IEEE 802.5):** Older LAN technology, uses a token-passing mechanism for media access. Devices form a logical ring.
- **FDDI (Fiber Distributed Data Interface):** High-speed token-passing LAN over fiber optics, uses dual rings for redundancy.
- **Wi-Fi (IEEE 802.11):** Wireless LAN technology, uses CSMA/CA (Collision Avoidance).

Common Pitfalls and Problem-Solving Techniques:

- **Pitfall:** Confusing the MAC access methods (CSMA/CD vs. Token Passing vs. CSMA/CA).
- **Technique:** Compare and contrast the characteristics, advantages, and disadvantages of different LAN technologies.

MAC Protocol (Medium Access Control)

MAC protocols define how multiple stations share a common transmission medium to avoid collisions and ensure fair access. They are part of the Data Link Layer.

Key Concepts:

- **Channel Partitioning Protocols:** Divide the channel into smaller, independent channels (e.g., TDMA, FDMA, CDMA).
- **Random Access Protocols:** Allow stations to transmit whenever they have data, resolving collisions if they occur (e.g., Aloha, CSMA, CSMA/CD, CSMA/CA).
- **Taking-Turns Protocols:** Stations take turns accessing the channel (e.g., Token Ring, Polling).

Important Formulas and Results:

- (See Pure Aloha, Slotted Aloha, CSMA/CD for specific formulas).

Common Pitfalls and Problem-Solving Techniques:

- **Pitfall:** Misidentifying the type of MAC protocol or its collision resolution mechanism.
- **Technique:** Understand the trade-offs between different MAC protocols (efficiency, fairness, complexity).

Network Flow

Network flow is a concept in graph theory used to model the movement of resources through a network. In computer networks, it can represent data traffic.

Key Concepts:

- **Flow Network:** A directed graph where each edge has a capacity and a flow.
- **Source and Sink:** Special nodes representing the origin and destination of the flow.
- **Max-Flow Min-Cut Theorem:** States that the maximum amount of flow passing from a source to a sink in a flow network is equal to the total capacity of the minimum cut.

Common Pitfalls and Problem-Solving Techniques:

- **Pitfall:** While Max-Flow Min-Cut is a core concept, GATE CS typically tests its conceptual understanding rather than complex numerical algorithms like Edmonds-Karp or Dinic.
- **Technique:** Focus on understanding the theorem and identifying cuts in simple networks.

Network Layer

The Network Layer (Layer 3 of the OSI model) is responsible for logical addressing, routing, and forwarding packets across different networks. IP is the primary protocol at this layer.

Key Concepts:

- **Logical Addressing:** IP addresses provide unique identification for devices across the internet.
- **Routing:** Determining the best path for packets to travel from source to destination.
- **Forwarding:** Moving a packet from an incoming interface to an outgoing interface on a router.
- **Protocols:** IP, ICMP, IGMP, Routing Protocols (RIP, OSPF, BGP).

Common Pitfalls and Problem-Solving Techniques:

- **Pitfall:** Confusing Network Layer functions with Data Link Layer (MAC addressing, framing) or Transport Layer (port numbers, reliability).
- **Technique:** Clearly distinguish the responsibilities of each layer in the OSI/TCP-IP model.

Network Protocols

Network protocols are formal rules and standards that govern how devices communicate and exchange data over a network. They define the format, timing, sequencing, and error control of data transmission.

Key Concepts:

- **TCP/IP Suite:** The most widely used set of protocols, forming the basis of the internet. Includes TCP, UDP, IP, HTTP, FTP, etc.
- **Layered Architecture:** Protocols are organized into layers (e.g., OSI model, TCP/IP model), with each layer providing services to the layer above it.

Common Pitfalls and Problem-Solving Techniques:

- **Pitfall:** Not knowing which protocols operate at which layer or their specific functions.
- **Technique:** Create a mental map or table of protocols and their corresponding layers/functions.

Network Switching

Network switching refers to the mechanisms used to establish connections between communicating devices in a network. Different switching techniques have varying characteristics regarding resource allocation and delay.

Key Concepts:

- **Circuit Switching:** A dedicated end-to-end communication path (circuit) is established before data transfer and maintained for the duration of the communication (e.g., traditional telephone networks). Guarantees bandwidth.
- **Packet Switching:** Data is broken into small, independent packets, each routed individually through the network. No dedicated path. More efficient use of bandwidth, but variable delay (e.g., Internet).
- **Message Switching:** Entire messages are stored and forwarded at each intermediate node. No dedicated path. High delay.

Common Pitfalls and Problem-Solving Techniques:

- **Pitfall:** Confusing the characteristics of circuit vs. packet switching, especially regarding setup time, resource reservation, and delay.
- **Technique:** Understand the trade-offs: Circuit switching for guaranteed QoS, Packet switching for efficiency and flexibility.

OSI Model

The OSI (Open Systems Interconnection) model is a conceptual framework that standardizes the functions of a telecommunication or computing system into seven distinct layers. It helps in understanding network architecture and troubleshooting.

Key Concepts:

1. **Physical Layer (Layer 1):** Deals with the physical transmission of raw bits over a medium (cables, connectors, voltage levels). PDU: Bit.
2. **Data Link Layer (Layer 2):** Provides reliable data transfer between adjacent nodes, framing, MAC addressing, error detection/correction. PDU: Frame.
3. **Network Layer (Layer 3):** Handles logical addressing (IP), routing, and forwarding packets across networks. PDU: Packet/Datagram.
4. **Transport Layer (Layer 4):** Provides end-to-end communication, reliability (TCP), flow control, congestion control, port addressing. PDU: Segment (TCP), Datagram (UDP).
5. **Session Layer (Layer 5):** Manages communication sessions, synchronization, dialog control.
6. **Presentation Layer (Layer 6):** Handles data representation, encryption, decryption, compression.
7. **Application Layer (Layer 7):** Provides network services directly to user applications (HTTP, FTP, DNS). PDU: Data/Message.

Common Pitfalls and Problem-Solving Techniques:

- **Pitfall:** Forgetting the order of layers or the primary function/PDU of each layer.
- **Technique:** Use mnemonics (e.g., "Please Do Not Throw Sausage Pizza Away") to remember the order. Focus on the first four layers, as they are most frequently tested.

Probability

Probability theory is used in computer networks to model and analyze the performance of random access protocols (like Aloha) and to evaluate network reliability and queueing behavior.

Key Concepts:

- **Poisson Distribution:** Often used to model arrival rates of packets or frames in random access protocols.
- **Throughput:** The rate at which successful transmissions occur.
- **Offered Load (G):** The total number of frames generated per unit of time, including retransmissions.

Important Formulas and Results:

- (See Pure Aloha, Slotted Aloha for specific probability-based throughput formulas).

Common Pitfalls and Problem-Solving Techniques:

- **Pitfall:** Misinterpreting the meaning of offered load (G) vs. throughput (S).

- **Technique:** Understand the assumptions behind probabilistic models (e.g., Poisson arrivals, infinite number of users).

Pure Aloha

Pure Aloha is a simple random access MAC protocol where stations transmit frames whenever they have data. If a collision occurs, the stations wait a random amount of time and retransmit.

Important Formulas and Results:

- **Throughput (S):** The rate of successful transmissions.

$$S = G \cdot e^{-2G}$$

  where G is the offered load (total number of frames generated per frame transmission time, including retransmissions).
- **Maximum Throughput (S_{max}):** Occurs when $G = 0.5$.

$$S_{max} = 0.5 \cdot e^{-2 \cdot 0.5} = 0.5 \cdot e^{-1} = \frac{1}{2e} \approx 0.184$$

Key Properties and Identities:

- Collision window is $2 \times T_{frame}$.
- Highly inefficient due to frequent collisions.

Common Pitfalls and Problem-Solving Techniques:

- **Pitfall:** Forgetting the factor of 2G in the exponent for pure Aloha.
- **Technique:** Understand the concept of the vulnerability period for collisions.

Routing

Routing is the process of selecting paths in a network along which to send network traffic. It involves determining the best route for packets from a source to a destination.

Key Concepts:

- **Static Routing:** Routes are manually configured by an administrator. Simple for small networks.
- **Dynamic Routing:** Routers exchange routing information and update their tables automatically using routing protocols. Adaptable to network changes.
- **Routing Table:** A table stored in a router that maps destination network addresses to the next hop router and outgoing interface.

Common Pitfalls and Problem-Solving Techniques:

- **Pitfall:** Confusing routing (path selection) with forwarding (packet movement).
- **Technique:** Understand how a router uses its routing table to make forwarding decisions.

Routing Protocols

Routing protocols are the algorithms and rules that routers use to exchange routing information and build their routing tables. They are classified by their operating scope and algorithm type.

Key Concepts:

- **Interior Gateway Protocols (IGPs):** Used within an Autonomous System (AS).
  - **RIP (Routing Information Protocol):** Distance Vector, uses hop count as metric, max 15 hops.
  - **OSPF (Open Shortest Path First):** Link State, uses Dijkstra's algorithm, builds a complete topology map.
- **Exterior Gateway Protocols (EGPs):** Used between Autonomous Systems.
  - **BGP (Border Gateway Protocol):** Path Vector, used for inter-domain routing on the Internet.

- **Link State vs. Distance Vector:**
  - **Distance Vector:** Routers share their entire routing tables with neighbors. (e.g., RIP)
  - **Link State:** Routers share information about their directly connected links with all other routers in the AS. (e.g., OSPF)

Common Pitfalls and Problem-Solving Techniques:

- **Pitfall:** Confusing RIP with OSPF, or IGPs with EGPs.
- **Technique:** Know the key characteristics of each protocol (metric, algorithm type, scope). Practice Dijkstra's algorithm for OSPF-like problems.

Sliding Window

Sliding Window protocols are flow control mechanisms that allow a sender to transmit multiple frames before receiving an acknowledgment, improving channel utilization. They use a window of sequence numbers.

Important Formulas and Results:

- **Window Size (W):** The maximum number of unacknowledged frames a sender can transmit.
- **Sequence Number Bits (n):** For a window size W, the minimum number of bits required for sequence numbers depends on the protocol.
  - **Go-Back-N:** Sender window W_S, Receiver window $W_R = 1$. Sequence numbers range from 0 to $2^n - 1$. $W_S + W_R \leq 2^n \implies W_S \leq 2^n - 1$.
  - **Selective Repeat:** Sender window W_S, Receiver window W_R. $W_S = W_R = 2^{n-1}$.
- **Channel Utilization (U) for Sliding Window:**

$$U = \min\left(1, \frac{W_S}{1+2a}\right)$$

  where $a = T_p/T_t$.

Key Properties and Identities:

- **Go-Back-N (GBN):** If a frame is lost, the sender retransmits that frame and all subsequent frames already sent. Receiver discards out-of-order frames.
- **Selective Repeat (SR):** If a frame is lost, only that specific frame is retransmitted. Receiver buffers out-of-order frames. More efficient but complex.

Common Pitfalls and Problem-Solving Techniques:

- **Pitfall:** Incorrectly determining the maximum window size for Go-Back-N vs. Selective Repeat for a given number of sequence bits.
- **Technique:** Memorize the window size rules for GBN and SR. Practice calculating utilization with different window sizes and a values.

Slotted Aloha

Slotted Aloha is an improvement over Pure Aloha where time is divided into discrete slots. Stations are only allowed to transmit at the beginning of a slot, reducing the collision window.

Important Formulas and Results:

- **Throughput (S):**

$$S = G \cdot e^{-G}$$

  where G is the offered load.
- **Maximum Throughput (S_{max}):** Occurs when $G = 1$.

$$S_{max} = 1 \cdot e^{-1} = \frac{1}{e} \approx 0.368$$

Key Properties and Identities:

- Collision window is $1 \times T_{frame}$.
- Requires global time synchronization.

Common Pitfalls and Problem-Solving Techniques:

- **Pitfall:** Confusing the throughput formula with Pure Aloha (exponent is -G, not -2G).
- **Technique:** Understand how slotting reduces the vulnerability period for collisions.

Sockets

Sockets provide an Application Programming Interface (API) for network communication, allowing applications to send and receive data over a network. They are the endpoints of communication.

Key Concepts:

- **Socket Types:**
  - **Stream Sockets (TCP):** Connection-oriented, reliable, ordered data delivery.
  - **Datagram Sockets (UDP):** Connectionless, unreliable, unordered data delivery.
- **Common Socket Functions (for TCP server):** socket(), bind(), listen(), accept(), send(), recv(), close().
- **Common Socket Functions (for TCP client):** socket(), connect(), send(), recv(), close().

Common Pitfalls and Problem-Solving Techniques:

- **Pitfall:** Confusing the sequence of system calls for client vs. server, or for TCP vs. UDP.
- **Technique:** Memorize the typical sequence of socket calls for both client and server applications.

Stop and Wait

Stop and Wait is the simplest ARQ (Automatic Repeat Request) protocol for reliable data transfer. The sender transmits one frame and waits for an acknowledgment (ACK) before sending the next.

Important Formulas and Results:

- **Transmission Time (T_t):** Time to put the entire frame on the link.

$$T_t = \frac{\text{Frame Size}}{\text{Bandwidth}}$$

- **Propagation Delay (T_p):** Time for the first bit to travel from sender to receiver.

$$T_p = \frac{\text{Distance}}{\text{Propagation Speed}}$$

- **Dimensionless Parameter (a):** Ratio of propagation delay to transmission time.

$$a = \frac{T_p}{T_t}$$

- **Channel Utilization (Efficiency) (U):**

$$U = \frac{1}{1 + 2a}$$

  This assumes no errors and negligible ACK transmission time.

Key Properties and Identities:

- Sender window size = 1, Receiver window size = 1.
- Uses 1-bit sequence numbers (0 and 1).
- Low utilization for high bandwidth-delay product links.

Common Pitfalls and Problem-Solving Techniques:

- **Pitfall:** Forgetting to multiply T_p by 2 for the round-trip propagation delay in the denominator of the utilization formula.
- **Technique:** Always calculate T_t and T_p first, then a, then U. Pay attention to units.

Subnetting

Subnetting is the process of dividing a larger IP network into smaller, more manageable subnetworks (subnets). It improves network efficiency, security, and reduces broadcast traffic.

Important Formulas and Results:

- **Number of Subnets:** If s bits are borrowed from the host portion of the IP address to create subnets, the number of subnets created is 2^s.
- **Number of Usable Hosts per Subnet:** If h bits remain for host IDs in a subnet, the number of usable hosts is $2^h - 2$ (excluding the network and broadcast addresses for that subnet).
- **Subnet Mask:** A 32-bit number that distinguishes the network portion from the host portion of an IP address. All network/subnet bits are 1, all host bits are 0.

Common Pitfalls and Problem-Solving Techniques:

- **Pitfall:** Incorrectly identifying the network ID, broadcast ID, or valid host range for a given IP address and subnet mask.
- **Technique:** Convert IP addresses and subnet masks to binary. Draw out the network, subnet, and host portions. Practice finding the first/last usable IP.

TCP (Transmission Control Protocol)

TCP is a reliable, connection-oriented, byte-stream Transport Layer protocol. It provides error control, flow control, and congestion control, making it suitable for applications requiring high data integrity.

Key Concepts:

- **Connection-Oriented:** Establishes a connection using a 3-way handshake before data transfer.
- **Reliable:** Uses sequence numbers, acknowledgments (ACKs), and retransmissions to ensure all data arrives correctly.
- **Flow Control:** Prevents a fast sender from overwhelming a slow receiver using a receive window (rwnd).
- **Congestion Control:** Prevents network congestion (see Congestion Control topic).
- **Full-Duplex:** Data can flow in both directions simultaneously.
- **Header Fields:** Source/Destination Port, Sequence Number, Acknowledgment Number, Window Size, Checksum, Flags (SYN, ACK, FIN, RST, PSH, URG).

Important Formulas and Results:

- **3-Way Handshake:** SYN -> SYN+ACK -> ACK.
- **4-Way Handshake (Connection Termination):** FIN -> ACK -> FIN -> ACK.

Common Pitfalls and Problem-Solving Techniques:

- **Pitfall:** Confusing sequence numbers with acknowledgment numbers, or the purpose of different TCP flags.
- **Technique:** Trace the sequence and acknowledgment numbers during a TCP connection setup, data transfer, and termination. Understand the role of the advertised window.

Token Bucket

The Token Bucket algorithm is a traffic shaping technique used to control the rate at which data is sent over a network. It smooths out bursty traffic and enforces a maximum average rate.

Key Concepts:

- **Bucket Capacity (C):** The maximum number of tokens the bucket can hold. Represents the maximum burst size allowed.
- **Token Arrival Rate (r):** The rate at which tokens are added to the bucket (e.g., tokens/second). Represents the average allowed transmission rate.
- **Traffic Shaping:** Packets can only be sent if there are enough tokens in the bucket. If not, packets are buffered or dropped.

Important Formulas and Results:

- **Maximum Output Rate:** Can be up to the link speed for a burst, but limited by the token rate r over time.
- **Maximum Burst Size:** Equal to the bucket capacity C.

Common Pitfalls and Problem-Solving Techniques:

- **Pitfall:** Confusing token bucket with leaky bucket. Token bucket allows bursts up to capacity, leaky bucket smooths all traffic to a constant rate.
- **Technique:** Calculate the maximum burst duration or the time it takes for the bucket to fill given a token rate and capacity.

UDP (User Datagram Protocol)

UDP is a simple, connectionless, unreliable Transport Layer protocol. It provides minimal services, primarily multiplexing/demultiplexing and basic error checking (optional checksum).

Key Concepts:

- **Connectionless:** No connection setup or teardown.
- **Unreliable:** No guarantees of delivery, order, or duplicate protection.
- **No Flow Control, No Congestion Control:** Applications must handle these if needed.
- **Header Fields:** Source Port, Destination Port, Length, Checksum (optional).

Key Properties and Identities:

- Lower overhead and faster than TCP.
- Suitable for applications where speed is more critical than reliability (e.g., streaming media, DNS, VoIP).

Common Pitfalls and Problem-Solving Techniques:

- **Pitfall:** Attributing TCP features (reliability, flow control) to UDP.
- **Technique:** Understand the trade-offs between UDP and TCP and identify appropriate applications for each.

Wrap Around Time

Wrap Around Time (WAT) refers to the time it takes for the sequence numbers used in a protocol (like TCP) to cycle through all possible values and return to the starting point. It's crucial for preventing ambiguity when old segments with duplicate sequence numbers arrive late.

Important Formulas and Results:

- If sequence numbers are in bytes and N is the number of bits for the sequence number field:

$$\text{Max Sequence Number} = 2^N$$

$$\text{Wrap Around Time (WAT)} = \frac{2^N \times \text{Maximum Segment Size (MSS)}}{\text{Bandwidth}}$$

(This formula might vary based on how sequence numbers are defined, e.g., per byte or per segment).
- A simpler form often seen:

$$\text{WAT} = \frac{\text{Total Sequence Number Space}}{\text{Data Rate}}$$

Key Properties and Identities:

- WAT should ideally be greater than the maximum segment lifetime (MSL) to prevent old segments from being misinterpreted.
- For TCP, the sequence number field is 32 bits.

Common Pitfalls and Problem-Solving Techniques:

- **Pitfall:** Forgetting to account for the units of sequence numbers (bytes vs. segments) and bandwidth.
- **Technique:** Ensure consistent units. Understand the relationship between WAT and MSL.

Quick Formula Reference

Channel Utilization & Throughput:

- **Stop-and-Wait ARQ:** $U = \frac{1}{1+2a}$ where $a = T_p/T_t$
- **Sliding Window ARQ (Go-Back-N/Selective Repeat):** $U = \min\left(1, \frac{W}{1+2a}\right)$
- **Pure Aloha Throughput:** $S = G \cdot e^{-2G}$ (Max $S = 1/(2e)$ at $G = 0.5$)
- **Slotted Aloha Throughput:** $S = G \cdot e^{-G}$ (Max $S = 1/e$ at $G = 1$)
- **Transmission Time:** $T_t = \text{Frame Size}/\text{Bandwidth}$
- **Propagation Delay:** $T_p = \text{Distance}/\text{Propagation Speed}$

Error Detection & Correction:

- **Hamming Code Parity Bits:** $2^p \geq m + p + 1$
- **CRC Codeword:** $T(x) = x^r \cdot M(x) + R(x)$

MAC Protocols:

- **CSMA/CD Slot Time:** $2 \times \text{Propagation Delay}$
- **CSMA/CD Minimum Frame Size:** $\text{Bandwidth} \times 2 \times \text{Propagation Delay}$

IP Fragmentation:

- **Fragment Offset:** $\text{New Offset} = \frac{\text{Original Offset} \times 8 + \text{Data Length of Fragment}}{8}$ (in 8-byte units)

IP Addressing & Subnetting:

- **Usable Hosts:** $2^h - 2$ (where h is host bits)
- **Number of Subnets:** 2^s (where s is subnet bits)

Sliding Window Sequence Numbers:

- **Go-Back-N Sender Window W_S:** $W_S \leq 2^n - 1$ (where n is sequence number bits)
- **Selective Repeat Sender/Receiver Window W_S, W_R:** $W_S = W_R = 2^{n-1}$

Wrap Around Time:

- **WAT (bytes):** $\frac{2^N \times \text{MSS}}{\text{Bandwidth}}$ (where N is sequence number bits)

Important Tips for GATE

1. **Master OSI/TCP-IP Layers:** Understand the function of each layer, the protocols operating at each layer, and the PDU (Protocol Data Unit) names. This is foundational and frequently tested.
2. **Practice Numerical Problems Extensively:** Focus on channel utilization, throughput, propagation/transmission delay calculations, IP addressing, subnetting, and fragmentation. Pay close attention to units (bits vs. bytes, Mbps vs. Kbps, ms vs. seconds).
3. **Differentiate Flow Control vs. Congestion Control:** Understand their distinct goals, mechanisms, and the protocols that implement them (e.g., TCP's window mechanisms for both).
4. **Know Protocol Headers:** Be familiar with the key fields in Ethernet, IP, TCP, and UDP headers. Questions often test the purpose or size of specific fields.
5. **Understand ARQ Protocols:** Clearly distinguish Stop-and-Wait, Go-Back-N, and Selective Repeat in terms of window sizes, retransmission strategies, and efficiency.
6. **Be Precise with IP Addressing and Subnetting:** Convert to binary if needed. Don't confuse network ID, broadcast ID, and usable host addresses. Practice CIDR notation.
7. **Read Questions Carefully:** Look for keywords like "negligible ACK time," "error-free channel," "half-duplex," or "full-duplex" as they significantly impact formula application.
8. **Time Management:** Some numerical problems can be lengthy. If stuck, make an educated guess (if no negative marking) or move on and return if time permits.

2.1 Application Layer Protocols (13)

✍ **Practice Tests:** Test 1 (15Q) Test 2 (4Q)

2.1.1 Application Layer Protocols: GATE CSE 2008 | Question: 14, ISRO2016-74

What is the maximum size of data that the application layer can pass on to the TCP layer below?

A. Any size
B. 2^{16} bytes - size of TCP header
C. 2^{16} bytes
D. 1500 bytes

gatecse-2008 easy computer-networks application-layer-protocols isro2016

Answer key☟

2.1.2 Application Layer Protocols: GATE CSE 2011 | Question: 4

Consider the different activities related to email.

- m1 : Send an email from mail client to mail server
- m2 : Download an email from mailbox server to a mail client
- m3 : Checking email in a web browser

Which is the application level protocol used in each activity?

A. m1 : HTTP m2 : SMTP m3 : POP
B. m1 : SMTP m2 : FTP m3 : HTTP
C. m1 : SMTP m2 : POP m3 : HTTP
D. m1 : POP m2 : SMTP m3 : IMAP

gatecse-2011 computer-networks application-layer-protocols easy

Answer key☟

2.1.3 Application Layer Protocols: GATE CSE 2012 | Question: 10

The protocol data unit (PDU) for the application layer in the Internet stack is:

A. Segment
B. Datagram
C. Message
D. Frame

gatecse-2012 computer-networks application-layer-protocols easy

Answer key☟

2.1.4 Application Layer Protocols: GATE CSE 2016 | Set 1 | Question: 25

Which of the following is/are example(s) of stateful application layer protocol?

i. HTTP
ii. FTP
iii. TCP
iv. POP3

A. (i) and (ii) only B. (ii) and (iii) only C. (ii) and (iv) only D. (iv) only

gatecse-2016-set1 computer-networks application-layer-protocols normal

Answer key

2.1.5 Application Layer Protocols: GATE CSE 2019 | Question: 16

Which of the following protocol pairs can be used to send and retrieve e-mails (in that order)?

A. IMAP, POP3 B. SMTP, POP3 C. SMTP, MIME D. IMAP, SMTP

gatecse-2019 computer-networks application-layer-protocols one-mark

Answer key

2.1.6 Application Layer Protocols: GATE CSE 2020 | Question: 25

Assume that you have made a request for a web page through your web browser to a web server. Initially the browser cache is empty. Further, the browser is configured to send HTTP requests in non-persistent mode. The web page contains text and five very small images.The minimum number of TCP connections required to display the web page completely in your browser is__________.

gatecse-2020 numerical-answers computer-networks application-layer-protocols one-mark

Answer key

2.1.7 Application Layer Protocols: GATE CSE 2022 | Question: 25

Consider the resolution of the domain name www.gate.org.in by a DNS resolver. Assume that no resource records are cached anywhere across the DNS servers and that iterative query mechanism is used in the resolution. The number of DNS query-response pairs involved in completely resolving the domain name is ________________.

gatecse-2022 numerical-answers computer-networks one-mark application-layer-protocols

Answer key

2.1.8 Application Layer Protocols: GATE CSE 2026 | Set 1 | Question: 9

Which of the following statements is/are true with respect to the interaction of a web browser with a web server using HTTP 1.1?

A. HTTP 1.1 facilitates downloading multiple objects of the same webpage over the same TCP connection, if the objects are stored in the same server
B. HTTP 1.1 facilitates downloading multiple objects of the same webpage over the same TCP connection, even if they are stored in different servers
C. HTTP 1.1 facilitates sending a request for downloading one object without waiting for a previously requested object to be downloaded completely
D. HTTP 1.1 facilitates downloading multiple webpages on the same server to be downloaded over a single TCP connection

gatecse-2026-set1 computer-networks application-layer-protocols multiple-selects one-mark

Answer key

2.1.9 Application Layer Protocols: GATE CSE 2026 | Set 2 | Question: 12

Which one of the following protocols may need to broadcast some of its messages?

A. SMTP B. FTP C. DHCP D. HTTP

gatecse-2026-set2 computer-networks application-layer-protocols one-mark

Answer key

2.1.10 Application Layer Protocols: GATE IT 2005 | Question: 25

Consider the three commands : PROMPT, HEAD and RCPT.
Which of the following options indicate a correct association of these commands with protocols where these are used?

A. HTTP, SMTP, FTP
B. FTP, HTTP, SMTP
C. HTTP, FTP, SMTP
D. SMTP, HTTP, FTP

gateit-2005 computer-networks application-layer-protocols normal

Answer key

2.1.11 Application Layer Protocols: GATE IT 2005 | Question: 77

Assume that "host1.mydomain.dom" has an IP address of $145.128.16.8$. Which of the following options would be most appropriate as a subsequence of steps in performing the reverse lookup of $145.128.16.8$? In the following options "NS" is an abbreviation of "nameserver".

A. Query a NS for the root domain and then NS for the "dom" domains
B. Directly query a NS for "dom" and then a NS for "mydomain.dom" domains
C. Query a NS for in-addr.arpa and then a NS for $128.145.$in-addr.arpa domains
D. Directly query a NS for $145.$in-addr.arpa and then a NS for $128.145.$in-addr.arpa domains

gateit-2005 computer-networks normal application-layer-protocols

Answer key

2.1.12 Application Layer Protocols: GATE IT 2006 | Question: 18

HELO and PORT, respectively, are commands from the protocols:

A. FTP and HTTP
B. TELNET and POP3
C. HTTP and TELNET
D. SMTP and FTP

gateit-2006 computer-networks application-layer-protocols normal

Answer key

2.1.13 Application Layer Protocols: GATE IT 2008 | Question: 20

Provide the best matching between the entries in the two columns given in the table below:

I.	Proxy Server	a.	Firewall
II.	Kazaa, DC++	b.	Caching
III.	Slip	c.	P2P
IV.	DNS	d.	PPP

A. I-a, II-d, III-c, IV-b
B. I-b, II-d, III-c, IV-a
C. I-a, II-c, III-d, IV-b
D. I-b, II-c, III-d, IV-a

gateit-2008 computer-networks normal application-layer-protocols

Answer key

2.2 Arp (1)

2.2.1 Arp: GATE CSE 2025 | Set 2 | Question: 6

Consider the following statements:

i. Address Resolution Protocol (ARP) provides a mapping from an IP address to the corresponding hardware (link-layer) address.
ii. A single TCP segment from a sender S to a receiver R cannot carry both data from S to R and acknowledgement for a segment from R to S

Which one of the following is CORRECT?

A. Both (i) and (ii) are TRUE
B. (i) is TRUE and (ii) is FALSE
C. (i) is FALSE and (ii) is TRUE
D. Both (i) and (ii) are FALSE

gatecse2025-set2 computer-networks tcp arp easy one-mark

Answer key☟

2.3 Bit Stuffing (2)

2.3.1 Bit Stuffing: GATE CSE 2014 | Set 3 | Question: 24

A bit-stuffing based framing protocol uses an 8-bit delimiter pattern of $01111110.$ If the output bit-string after stuffing is $01111100101,$ then the input bit-string is:

A. 0111110100 B. 0111110101 C. 0111111101 D. 0111111111

gatecse-2014-set3 computer-networks error-detection bit-stuffing

Answer key☟

2.3.2 Bit Stuffing: GATE IT 2004 | Question: 80

In a data link protocol, the frame delimiter flag is given by $0111.$ Assuming that bit stuffing is employed, the transmitter sends the data sequence 01110110 as:

A. 01101011 B. 011010110 C. 011101100 D. 0110101100

gateit-2004 computer-networks network-flow normal bit-stuffing

Answer key☟

2.4 Bridges (3)

2.4.1 Bridges: GATE CSE 2004 | Question: 16

Which of the following is NOT true with respect to a transparent bridge and a router?

A. Both bridge and router selectively forward data packets
B. A bridge uses IP addresses while a router uses MAC addresses
C. A bridge builds up its routing table by inspecting incoming packets
D. A router can connect between a LAN and a WAN

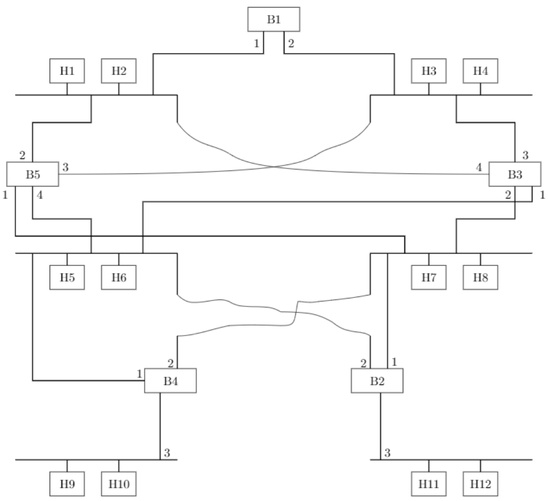
gatecse-2004 computer-networks bridges normal

Answer key☟

2.4.2 Bridges: GATE CSE 2006 | Question: 82

Consider the diagram shown below where a number of LANs are connected by (transparent) bridges. In order to avoid packets looping through circuits in the graph, the bridges organize themselves in a spanning tree. First, the root bridge is identified as the bridge with the least serial number. Next, the root sends out (one or more) data units to enable the setting up of the spanning tree of shortest paths from the root bridge to each bridge.

Each bridge identifies a port (the root port) through which it will forward frames to the root bridge. Port conflicts are always resolved in favour of the port with the lower index value. When there is a possibility of multiple bridges forwarding to the same LAN (but not through the root port), ties are broken as follows: bridges closest to the root get preference and between such bridges, the one with the lowest serial number is preferred.



For the given connection of LANs by bridges, which one of the following choices represents the depth first traversal of the spanning tree of bridges?

A. $B1, B5, B3, B4, B2$
B. $B1, B3, B5, B2, B4$
C. $B1, B5, B2, B3, B4$
D. $B1, B3, B4, B5, B2$

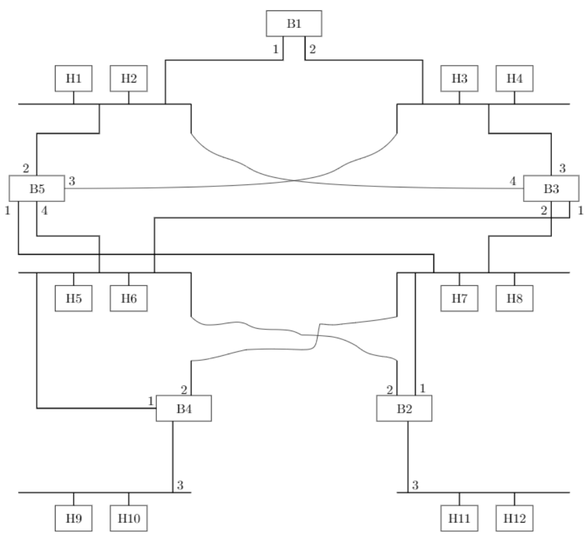
gatecse-2006 computer-networks bridges normal

Answer key

2.4.3 Bridges: GATE CSE 2006 | Question: 83

Consider the diagram shown below where a number of LANs are connected by (transparent) bridges. In order to avoid packets looping through circuits in the graph, the bridges organize themselves in a spanning tree. First, the root bridge is identified as the bridge with the least serial number. Next, the root sends out (one or more) data
units to enable the setting up of the spanning tree of shortest paths from the root bridge to each bridge.

Each bridge identifies a port (the root port) through which it will forward frames to the root bridge. Port conflicts are always resolved in favour of the port with the lower index value. When there is a possibility of multiple bridges forwarding to the same LAN (but not through the root port), ties are broken as follows: bridges closest to the root get preference and between such bridges, the one with the lowest serial number is preferred.



Consider the spanning tree $B1, B5, B3, B4, B2$ for the given connection of LANs by bridges, that represents the depth first traversal of the spanning tree of bridges. Let host $H1$ send out a broadcast ping packet. Which of the following options represents the correct forwarding table on $B3$?

a.

Hosts	Port
H1, H2, H3, H4	3
H5, H6, H9, H10	1
H7, H8, H11, H12	2

b.

Hosts	Port
H1, H2	4
H3, H4	3
H5, H6	1
H7, H8, H9, H10, H11, H12	2

c.

Hosts	Port
H3, H4	3
H5, H6, H9, H10	1
H1, H2	4
H7, H8, H11, H12	2

d.

Hosts	Port
H1, H2, H3, H4	3
H5, H7, H9, H10	1
H7, H8, H11, H12	4

gatecse-2006 computer-networks bridges normal

Answer key

2.5 CRC Polynomial (5)

Practice Test: Test 1 (7Q)

2.5.1 CRC Polynomial: GATE CSE 2007 | Question: 68, ISRO2016-73

The message 11001001 is to be transmitted using the CRC polynomial $x^3 + 1$ to protect it from errors. The message that should be transmitted is:

A. 11001001000 B. 11001001011 C. 11001010 D. 110010010011

gatecse-2007 computer-networks error-detection crc-polynomial normal isro2016

Answer key

2.5.2 CRC Polynomial: GATE CSE 2017 | Set 1 | Question: 32

A computer network uses polynomials over $GF(2)$ for error checking with 8 bits as information bits and uses $x^3 + x + 1$ as the generator polynomial to generate the check bits. In this network, the message 01011011 is transmitted as:

A. 01011011010 B. 01011011011 C. 01011011101 D. 01011011100

gatecse-2017-set1 computer-networks crc-polynomial normal

Answer key

2.5.3 CRC Polynomial: GATE CSE 2021 | Set 2 | Question: 34

Consider the cyclic redundancy check (CRC) based error detecting scheme having the generator polynomial $X^3 + X + 1$. Suppose the message $m_4m_3m_2m_1m_0 = 11000$ is to be transmitted. Check bits $c_2c_1c_0$ are appended at the end of the message by the transmitter using the above CRC scheme. The transmitted bit string is denoted by $m_4m_3m_2m_1m_0c_2c_1c_0$. The value of the checkbit sequence $c_2c_1c_0$ is

A. 101 B. 110 C. 100 D. 111

gatecse-2021-set2 computer-networks crc-polynomial two-marks

Answer key

2.5.4 CRC Polynomial: GATE CSE 2026 | Set 2 | Question: 33

Consider the transmission of data bits 110001011 over a link that uses Cyclic Redundancy Check (CRC) code for error detection. If the generator bit pattern is given to be 1001, which one of the following options shows the remainder bit pattern appended to the data bits before transmission?

A. 011 B. 101 C. 000 D. 100

gatecse-2026-set2 computer-networks crc-polynomial error-detection two-marks

Answer key

2.5.5 CRC Polynomial: GATE IT 2005 | Question: 78

Consider the following message $M = 1010001101$. The cyclic redundancy check (CRC) for this message using the divisor polynomial $x^5 + x^4 + x^2 + 1$ is :

A. 01110 B. 01011 C. 10101 D. 10110

gateit-2005 computer-networks crc-polynomial normal

Answer key

2.6 CSMA CD (6)

✍ **Practice Test:** Test 1 (10Q)

2.6.1 CSMA CD: GATE CSE 2015 | Set 3 | Question: 6

Consider a CSMA/CD network that transmits data at a rate of 100 **Mbps** $(10^8$ bits per second) over a 1 **km** (kilometre) cable with no repeaters. If the minimum frame size required for this network is 1250 bytes, What is the signal speed $(\textbf{km}/\textbf{sec})$ in the cable?

A. 8000 B. 10000 C. 16000 D. 20000

gatecse-2015-set3 computer-networks congestion-control csma-cd normal

Answer key

2.6.2 CSMA CD: GATE CSE 2016 | Set 2 | Question: 53

A network has a data transmission bandwidth of 20×10^6 bits per second. It uses **CSMA/CD** in the **MAC** layer. The maximum signal propagation time from one node to another node is 40 microseconds. The minimum size of a frame in the network is __________ bytes.

gatecse-2016-set2 computer-networks csma-cd numerical-answers normal

Answer key

2.6.3 CSMA CD: GATE CSE 2018 | Question: 55

Consider a simple communication system where multiple nodes are connected by a shared broadcast medium (like Ethernet or wireless). The nodes in the system use the following carrier-sense based medium access protocol. A node that receives a packet to transmit will carrier-sense the medium for 5 units of time. If the node does not detect any other transmission, it starts transmitting its packet in the next time unit. If the node detects another transmission, it waits until this other transmission finishes, and then begins to carrier-sense for 5 time units again. Once they start to transmit, nodes do not perform any collision detection and continue transmission even if a collision occurs. All transmissions last for 20 units of time. Assume that the transmission signal travels at the speed of 10 meters per unit time in the medium.

Assume that the system has two nodes P and Q, located at a distance d meters from each other. P start transmitting a packet at time $t = 0$ after successfully completing its carrier-sense phase. Node Q has a packet to transmit at time $t = 0$ and begins to carrier-sense the medium.

The maximum distance d (in meters, rounded to the closest integer) that allows Q to successfully avoid a collision between its proposed transmission and P's ongoing transmission is _______.

gatecse-2018 computer-networks csma-cd numerical-answers two-marks

Answer key

2.6.4 CSMA CD: GATE IT 2005 | Question: 27

Which of the following statements is TRUE about CSMA/CD:

A. IEEE 802.11 wireless LAN runs CSMA/CD protocol
B. Ethernet is not based on CSMA/CD protocol
C. CSMA/CD is not suitable for a high propagation delay network like satellite network
D. There is no contention in a CSMA/CD network

gateit-2005 computer-networks congestion-control csma-cd normal

Answer key

2.6.5 CSMA CD: GATE IT 2005 | Question: 71

A network with CSMA/CD protocol in the MAC layer is running at 1Gbps over a 1km cable with no repeaters. The signal speed in the cable is $2 \times 10^8 \text{m/sec}$. The minimum frame size for this network should be:

A. 10000bits
B. 10000bytes
C. 5000 bits
D. 5000bytes

gateit-2005 computer-networks congestion-control csma-cd normal

Answer key

2.6.6 CSMA CD: GATE IT 2008 | Question: 65

The minimum frame size required for a CSMA/CD based computer network running at 1Gbps on a $200m$ cable with a link speed of $2 \times 10^8 \text{m/sec}$ is:

A. 125bytes
B. 250bytes
C. 500bytes
D. None of the above

gateit-2008 computer-networks csma-cd normal

Answer key

2.7 Channel Utilization (1)

2.7.1 Channel Utilization: GATE CSE 2025 | Set 2 | Question: 26

Suppose we are transmitting frames between two nodes using Stop-and-Wait protocol. The frame size is 3000 bits. The transmission rate of the channel is 2000 bps (bits/second) and the propagation delay between the two nodes is 100 milliseconds. Assume that the processing times at the source and destination are negligible. Also, assume that the size of the acknowledgement packet is negligible. Which ONE of the following most accurately gives the channel utilization for the above scenario in percentage?

A. 88.23
B. 93.75
C. 85.44
D. 66.67

gatecse2025-set2 computer-networks stop-and-wait channel-utilization two-marks

Answer key

2.8 Communication (4)

✍ **Practice Test:** Test 1 (11Q)

2.8.1 Communication: GATE CSE 2012 | Question: 44

Consider a source computer (S) transmitting a file of size 10^6 bits to a destination computer (D) over a network of two routers (R_1 and R_2) and three links ($L_1, L_2,$ and L_3). L_1 connects S to R_1; L_2 connects R_1 to R_2; and L_3 connects R_2 to D. Let each link be of length 100 km. Assume signals travel over each link at a speed of 10^8 meters per second. Assume that the link bandwidth on each link is 1 Mbps. Let the file be broken down into 1000 packets each of size 1000 bits. Find the total sum of transmission and propagation delays in transmitting the file from S to D?

A. 1005 ms B. 1010 ms C. 3000 ms D. 3003 ms

gatecse-2012 computer-networks communication normal

Answer key

2.8.2 Communication: GATE CSE 2022 | Question: 49

Consider a 100 Mbps link between an earth station (sender) and a satellite (receiver) at an altitude of 2100 km. The signal propagates at a speed of 3×10^8 m/s. The time taken (in milliseconds, *rounded off to two decimal places*) for the receiver to completely receive a packet of 1000 bytes transmitted by the sender is _______.

gatecse-2022 numerical-answers computer-networks two-marks communication

Answer key

2.8.3 Communication: GATE IT 2007 | Question: 62

Let us consider a statistical time division multiplexing of packets. The number of sources is 10. In a time unit, a source transmits a packet of 1000 bits. The number of sources sending data for the first 20 time units is $6, 9, 3, 7, 2, 2, 2, 3, 4, 6, 1, 10, 7, 5, 8, 3, 6, 2, 9, 5$ respectively. The output capacity of multiplexer is 5000 bits per time unit. Then the average number of backlogged of packets per time unit during the given period is

A. 5 B. 4.45 C. 3.45 D. 0

gateit-2007 computer-networks communication normal

Answer key

2.8.4 Communication: GATE IT 2007 | Question: 64

A broadcast channel has 10 nodes and total capacity of 10 Mbps. It uses polling for medium access. Once a node finishes transmission, there is a polling delay of 80 μs to poll the next node. Whenever a node is polled, it is allowed to transmit a maximum of 1000 bytes. The maximum throughput of the broadcast channel is:

A. 1 Mbps B. $100/11$ Mbps C. 10 Mbps D. 100 Mbps

gateit-2007 computer-networks communication normal

Answer key

2.9 Congestion Control (9)

✍ **Practice Test:** Test 1 (14Q)

2.9.1 Congestion Control: GATE CSE 2008 | Question: 56

In the slow start phase of the TCP congestion algorithm, the size of the congestion window:

A. does not increase
B. increase linearly
C. increases quadratically
D. increases exponentially

gatecse-2008 computer-networks congestion-control normal

Answer key

2.9.2 Congestion Control: GATE CSE 2012 | Question: 45

Consider an instance of TCP's Additive Increase Multiplicative Decrease (AIMD) algorithm where the window size at the start of the slow start phase is 2 MSS and the threshold at the start of the first transmission is 8 MSS. Assume that a timeout occurs during the fifth transmission. Find the congestion window size at the end of the tenth transmission.

A. 8 MSS B. 14 MSS C. 7 MSS D. 12 MSS

gatecse-2012 computer-networks congestion-control normal

Answer key

2.9.3 Congestion Control: GATE CSE 2014 | Set 1 | Question: 27

Let the size of congestion window of a TCP connection be 32 KB when a timeout occurs. The round trip time of the connection is 100 msec and the maximum segment size used is 2 KB. The time taken **(in msec)** by the TCP connection to get back to 32 KB congestion window is _________.

gatecse-2014-set1 computer-networks tcp congestion-control numerical-answers normal

Answer key

2.9.4 Congestion Control: GATE CSE 2018 | Question: 14

Consider the following statements regarding the slow start phase of the TCP congestion control algorithm. Note that *cwnd* stands for the TCP congestion window and MSS window denotes the Maximum Segments Size:

i. The *cwnd* increases by 2 MSS on every successful acknowledgment
ii. The *cwnd* approximately doubles on every successful acknowledgment
iii. The *cwnd* increases by 1 MSS every round trip time
iv. The *cwnd* approximately doubles every round trip time

Which one of the following is correct?

A. Only (ii) and (iii) are true
B. Only (i) and (iii) are true
C. Only (iv) is true
D. Only (i) and (iv) are true

gatecse-2018 computer-networks tcp congestion-control normal one-mark

Answer key

2.9.5 Congestion Control: GATE CSE 2020 | Question: 55

Consider a TCP connection between a client and a server with the following specifications; the round trip time is 6 ms, the size of the receiver advertised window is 50 KB, slow-start threshold at the client is 32 KB, and the maximum segment size is 2 KB. The connection is established at time $t=0$. Assume that there are no timeouts and errors during transmission. Then the size of the congestion window (in KB) at time $t+60$ ms after all acknowledgements are processed is _______

gatecse-2020 numerical-answers computer-networks tcp congestion-control two-marks

Answer key

2.9.6 Congestion Control: GATE CSE 2024 | Set 2 | Question: 44

Consider a TCP connection operating at a point of time with the congestion window of size 12 MSS (Maximum Segment Size), when a timeout occurs due to packet loss. Assuming that all the segments transmitted in the next two RTTs (Round Trip Time) are acknowledged correctly, the congestion window size (*in* MSS) during the third RTT will be __________.

gatecse-2024-set2 numerical-answers computer-networks tcp congestion-control two-marks

Answer key

2.9.7 Congestion Control: GATE CSE 2026 | Set 1 | Question: 34

A TCP sender successfully establishes a connection with a TCP receiver and starts the transmission of segments. The TCP congestion control mechanism's slow-start threshold is set to 10000 segments. Assume that the round-trip time is fixed at 1 millisecond. Assume that the sender always has data to send, the segments are numbered from 1, and no segment is lost. Let t denote the time (in milliseconds) at which the transmission of segment number 2000 starts.

Which one of the following options is correct?

A. $9 \leq t < 10$
B. $10 \leq t < 11$
C. $11 \leq t < 12$
D. $12 \leq t < 13$

gatecse-2026-set1 two-marks computer-networks congestion-control tcp

Answer key

2.9.8 Congestion Control: GATE CSE 2026 | Set 2 | Question: 48

Consider a new TCP connection between a sender and a receiver. The receiver advertised window is constant at 48 KB, the maximum segment size (MSS) is 2 KB, and the slow start threshold for TCP congestion control is 16 KB. Assume that there are no timeouts or duplicate acknowledgements. The number of rounds of transmission required for the congestion control algorithm of the TCP connection to reach the congestion avoidance phase is _____. (answer in integer)

Note: $1 \text{ K} = 2^{10}$

gatecse-2026-set2 computer-networks congestion-control tcp numerical-answers two-marks

Answer key

2.9.9 Congestion Control: GATE IT 2005 | Question: 73

On a TCP connection, current congestion window size is Congestion Window = 4 KB. The window size advertised by the receiver is Advertise Window = 6 KB. The last byte sent by the sender is LastByteSent = 10240 and the last byte acknowledged by the receiver is LastByteAcked = 8192. The current window size at the sender is:

A. 2048 bytes
B. 4096 bytes
C. 6144 bytes
D. 8192 bytes

gateit-2005 computer-networks congestion-control normal

Answer key

2.10 Data Communication (1)

2.10.1 Data Communication: GATE CSE 2026 | Set 2 | Question: 55

It is necessary to design a link-layer protocol between two hosts that are directly connected over a lossless link of length 3000 kilometers. Assume that the link bandwidth is 10^8 bits per second and that the propagation delay in the link is 5 nanoseconds per meter. Every transmitted data byte is assigned a unique sequence number.

Let N be the minimum number of bits needed for the sequence number field in the protocol header such that

i. the sequence numbers do not wrap around before 60 seconds, and
ii. the maximum utilization of the link is achieved.

The value of N is _____ . (answer in integer)

gatecse-2026-set2 computer-networks data-communication numerical-answers two-marks

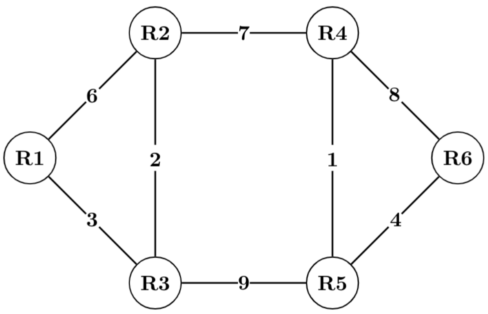
Answer key

2.11 Distance Vector Routing (8)

✍ **Practice Test:** Test 1 (15Q)

2.11.1 Distance Vector Routing: GATE CSE 2010 | Question: 54

Consider a network with 6 routers $\mathbf{R1}$ to $\mathbf{R6}$ connected with links having weights as shown in the following diagram.



All the routers use the distance vector based routing algorithm to update their routing tables. Each router starts with its routing table initialized to contain an entry for each neighbor with the weight of the respective connecting link. After all the routing tables stabilize, how many links in the network will never be used for carrying any data?

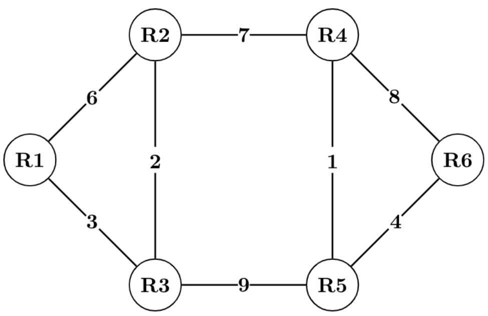
A. 4 B. 3 C. 2 D. 1

gatecse-2010 computer-networks routing distance-vector-routing normal

Answer key☟

2.11.2 Distance Vector Routing: GATE CSE 2010 | Question: 55

Consider a network with 6 routers $R1$ to $R6$ connected with links having weights as shown in the following diagram.



Suppose the weights of all unused links are changed to 2 and the distance vector algorithm is used again until all routing tables stabilize. How many links will now remain unused?

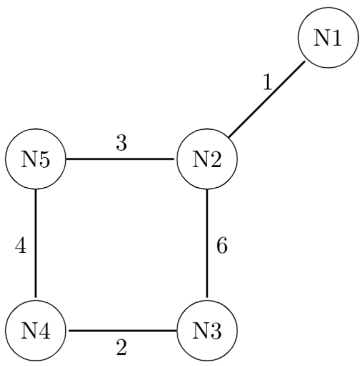
A. 0 B. 1 C. 2 D. 3

gatecse-2010 computer-networks routing distance-vector-routing normal

Answer key☟

2.11.3 Distance Vector Routing: GATE CSE 2011 | Question: 52

Consider a network with five nodes, $N1$ to $N5$, as shown as below.



The network uses a Distance Vector Routing protocol. Once the routes have been stabilized, the distance vectors at different nodes are as follows.

N1: $(0, 1, 7, 8, 4)$

N2: $(1, 0, 6, 7, 3)$

N3: $(7, 6, 0, 2, 6)$

N4: $(8, 7, 2, 0, 4)$

N5: $(4, 3, 6, 4, 0)$

Each distance vector is the distance of the best known path at that instance to nodes, $N1$ to $N5$, where the distance to itself is 0. Also, all links are symmetric and the cost is identical in both directions. In each round, all nodes exchange their distance vectors with their respective neighbors. Then all nodes update their distance vectors. In between two rounds, any change in cost of a link will cause the two incident nodes to change only that entry in their distance vectors.

The cost of link $N2 - N3$ reduces to 2 (in both directions). After the next round of updates, what will be the new distance vector at node, $N3$?

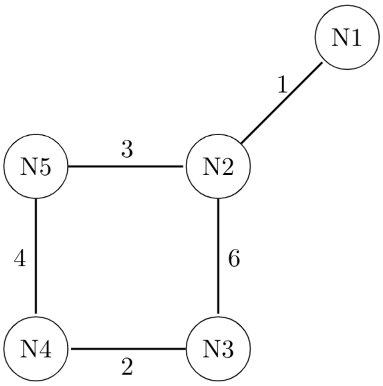
A. $(3,2,0,2,5)$
B. $(3,2,0,2,6)$
C. $(7,2,0,2,5)$
D. $(7,2,0,2,6)$

gatecse-2011 computer-networks routing distance-vector-routing normal

Answer key

2.11.4 Distance Vector Routing: GATE CSE 2011 | Question: 53

Consider a network with five nodes, $N1$ to $N5$, as shown as below.



The network uses a Distance Vector Routing protocol. Once the routes have been stabilized, the distance vectors at different nodes are as follows.

- **N1**: $(0, 1, 7, 8, 4)$
- **N2**: $(1, 0, 6, 7, 3)$
- **N3**: $(7, 6, 0, 2, 6)$
- **N4**: $(8, 7, 2, 0, 4)$
- **N5**: $(4, 3, 6, 4, 0)$

Each distance vector is the distance of the best known path at that instance to nodes, $N1$ to $N5$, where the distance to itself is 0. Also, all links are symmetric and the cost is identical in both directions. In each round, all nodes exchange their distance vectors with their respective neighbors. Then all nodes update their distance vectors. In between two rounds, any change in cost of a link will cause the two incident nodes to change only that entry in their distance vectors.

The cost of link $N2 - N3$ reduces to 2 (in both directions). After the next round of updates, the link $N1 - N2$ goes down. $N2$ will reflect this change immediately in its distance vector as cost, ∞. After the **NEXT ROUND** of update, what will be the cost to $N1$ in the distance vector of $N3$?

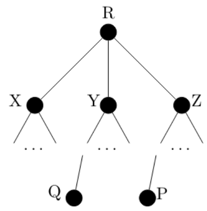
A. 3
B. 9
C. 10
D. ∞

gatecse-2011 computer-networks routing distance-vector-routing normal

Answer key

2.11.5 Distance Vector Routing: GATE CSE 2021 | Set 2 | Question: 45

Consider a computer network using the distance vector routing algorithm in its network layer. The partial topology of the network is shown below.



The objective is to find the shortest-cost path from the router R to routers P and Q. Assume that R does not initially know the shortest routes to P and Q. Assume that R has three neighbouring routers denoted as X, Y and Z. During one iteration, R measures its distance to its neighbours X, Y, and Z as 3, 2 and 5, respectively. Router R gets routing vectors from its neighbours that indicate that the distance to router P from routers X, Y and Z are 7, 6 and 5, respectively. The routing vector also indicates that the distance to router Q from routers X, Y and Z are 4, 6 and 8 respectively. Which of the following statement(s) is/are correct with respect to the new routing table o R, after updation during this iteration?

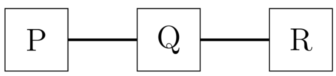
A. The distance from R to P will be stored as 10
B. The distance from R to Q will be stored as 7
C. The next hop router for a packet from R to P is Y
D. The next hop router for a packet from R to Q is Z

gatecse-2021-set2 multiple-selects computer-networks distance-vector-routing two-marks

Answer key☟

2.11.6 Distance Vector Routing: GATE CSE 2022 | Question: 47

Consider a network with three routers $\mathrm{P}, \mathrm{Q}, \mathrm{R}$ shown in the figure below. All the links have cost of unity.



The routers exchange distance vector routing information and have converged on the routing tables, after which the link Q-R fails. Assume that P and Q send out routing updates at random times, each at the same average rate. The probability of a routing loop formation *(rounded off to one decimal place)* between P and Q, leading to count-to-infinity problem, is _______________.

gatecse-2022 numerical-answers computer-networks routing distance-vector-routing two-marks

Answer key☟

2.11.7 Distance Vector Routing: GATE IT 2005 | Question: 29

Count to infinity is a problem associated with:

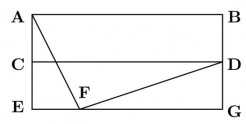
A. link state routing protocol.
B. distance vector routing protocol
C. DNS while resolving host name
D. TCP for congestion control

gateit-2005 computer-networks routing distance-vector-routing normal

Answer key☟

2.11.8 Distance Vector Routing: GATE IT 2007 | Question: 60

For the network given in the figure below, the routing tables of the four nodes A, E, D and G are shown. Suppose that F has estimated its delay to its neighbors, A, E, D and G as 8, 10, 12 and 6 msecs respectively and updates its routing table using distance vector routing technique.



Routing Table of A

A	0
B	40
C	14
D	17
E	21
F	9
G	24

Routing Table of D

A	20
B	8
C	30
D	0
E	14
F	7
G	22

Routing Table of E

A	24
B	27
C	7
D	20
E	0
F	11
G	22

Routing Table of G

A	21
B	24
C	22
D	19
E	22
F	10
G	0

A.

A	8
B	20
C	17
D	12
E	10
F	0
G	6

B.

A	21
B	8
C	7
D	19
E	14
F	0
G	22

C.

A	8
B	20
C	17
D	12
E	10
F	16
G	6

D.

A	8
B	8
C	7
D	12
E	10
F	0
G	6

gateit-2007 computer-networks distance-vector-routing normal

Answer key

2.12 Error Detection (8)

✍ **Practice Test:** Test 1 (9Q)

2.12.1 Error Detection: GATE CSE 1992 | Question: 01,ii

Consider a 3-bit error detection and 1-bit error correction hamming code for 4-bit data. The extra parity bits required would be _____ and the 3-bit error detection is possible because the code has a minimum distance of _______.

gate1992 computer-networks error-detection normal fill-in-the-blanks

Answer key

2.12.2 Error Detection: GATE CSE 1995 | Question: 1.12

What is the distance of the following code $000000, 010101, 000111, 011001, 111111$?

A. 2
B. 3
C. 4
D. 1

gate1995 computer-networks error-detection normal

Answer key

2.12.3 Error Detection: GATE CSE 2009 | Question: 48

Let $G(x)$ be the generator polynomial used for CRC checking. What is the condition that should be satisfied by $G(x)$ to detect odd number of bits in error?

A. $G(x)$ contains more than two terms
B. $G(x)$ does not divide $1+x^k$, for any k not exceeding the frame length
C. $1+x$ is a factor of $G(x)$
D. $G(x)$ has an odd number of terms.

gatecse-2009 computer-networks error-detection normal

Answer key

2.12.4 Error Detection: GATE CSE 2017 | Set 2 | Question: 34

Consider the binary code that consists of only four valid codewords as given below:

$00000, 01011, 10101, 11110$

Let the minimum Hamming distance of the code p and the maximum number of erroneous bits that can be corrected by the code be q. Then the values of p and q are

A. $p = 3$ and $q = 1$ B. $p = 3$ and $q = 2$ C. $p = 4$ and $q = 1$ D. $p = 4$ and $q = 2$

gatecse-2017-set2 computer-networks error-detection

Answer key

2.12.5 Error Detection: GATE IT 2005 | Question: 74

In a communication network, a packet of length L bits takes link L_1 with a probability of p_1 or link L_2 with a probability of p_2. Link L_1 and L_2 have bit error probability of b_1 and b_2 respectively. The probability that the packet will be received without error via either L_1 or L_2 is

A. $(1-b_1)^L p_1 + (1-b_2)^L p_2$
B. $[1-(b_1+b_2)^L]p_1 p_2$
C. $(1-b_1)^L (1-b_2)^L p_1 p_2$
D. $1-(b_1^L p_1 + b_2^L p_2)$

gateit-2005 computer-networks error-detection probability normal

Answer key

2.12.6 Error Detection: GATE IT 2007 | Question: 43

An error correcting code has the following code words: $00000000, 00001111, 01010101, 10101010, 11110000$. What is the maximum number of bit errors that can be corrected?

A. 0 B. 1 C. 2 D. 3

gateit-2007 computer-networks error-detection normal

Answer key

2.12.7 Error Detection: GATE IT 2008 | Question: 66

Data transmitted on a link uses the following $2D$ parity scheme for error detection:
Each sequence of 28 bits is arranged in a 4×7 matrix (rows r_0 through r_3, and columns d_7 through d_1) and is padded with a column d_0 and row r_4 of parity bits computed using the Even parity scheme. Each bit of column d_0 (respectively, row r_4) gives the parity of the corresponding row (respectively, column). These 40 bits are transmitted over the data link.

	$\mathbf{d_7}$	$\mathbf{d_6}$	$\mathbf{d_5}$	$\mathbf{d_4}$	$\mathbf{d_3}$	$\mathbf{d_2}$	$\mathbf{d_1}$	$\mathbf{d_0}$
$\mathbf{r_0}$	0	1	0	1	0	0	1	**1**
$\mathbf{r_1}$	1	1	0	0	1	1	1	**0**
$\mathbf{r_2}$	0	0	0	1	0	1	0	**0**
$\mathbf{r_3}$	0	1	1	0	1	0	1	**0**
$\mathbf{r_4}$	**1**	**1**	**0**	**0**	**0**	**1**	**1**	**0**

The table shows data received by a receiver and has n corrupted bits. What is the minimum possible value of n?

A. 1 B. 2 C. 3 D. 4

gateit-2008 computer-networks normal error-detection

Answer key

2.12.8 Error Detection: GATE1987-2-i

Match the pairs in the following questions:

(A) Cyclic Redundancy Code	(p) Error Correction
(B) Serial Communication	(q) Wired-OR
(C) Open Collector	(r) Error detection
(D) Hamming Code	(s) RS-232-C

gate1989 descriptive computer-networks error-detection

Answer key

2.13 Ethernet (7)

Practice Test: Test 1 (8Q)

2.13.1 Ethernet: GATE CSE 2004 | Question: 54

A and B are the only two stations on an Ethernet. Each has a steady queue of frames to send. Both A and B attempt to transmit a frame, collide, and A wins the first backoff race. At the end of this successful transmission by A, both A and B attempt to transmit and collide. The probability that A wins the second backoff race is:

A. 0.5
B. 0.625
C. 0.75
D. 1.0

gatecse-2004 computer-networks ethernet probability normal

Answer key

2.13.2 Ethernet: GATE CSE 2013 | Question: 36

Determine the maximum length of the cable (in km) for transmitting data at a rate of 500 Mbps in an Ethernet LAN with frames of size $10,000$ bits. Assume the signal speed in the cable to be $2,00,000$ km/s.

A. 1
B. 2
C. 2.5
D. 5

gatecse-2013 computer-networks ethernet normal

Answer key

2.13.3 Ethernet: GATE CSE 2016 | Set 2 | Question: 24

In an Ethernet local area network, which one of the following statements is **TRUE**?

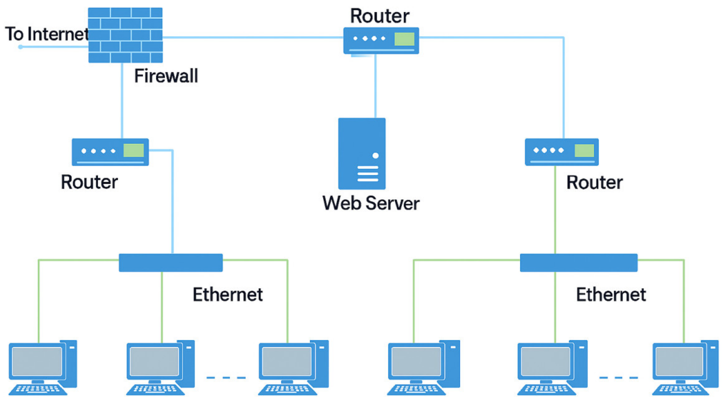
A. A station stops to sense the channel once it starts transmitting a frame.
B. The purpose of the jamming signal is to pad the frames that are smaller than the minimum frame size.
C. A station continues to transmit the packet even after the collision is detected.
D. The exponential back off mechanism reduces the probability of collision on retransmissions.

gatecse-2016-set2 computer-networks ethernet normal

Answer key

2.13.4 Ethernet: GATE CSE 2022 | Question: 12

Consider an enterprise network with two Ethernet segments, a web server and a firewall, connected via three routers as shown below.



What is the number of subnets inside the enterprise network?

A. 3 B. 12 C. 6 D. 8

gatecse-2022 computer-networks ethernet one-mark

Answer key

2.13.5 Ethernet: GATE CSE 2024 | Set 2 | Question: 13

Node X has a TCP connection open to node Y. The packets from X to Y go through an intermediate IP router R. Ethernet switch S is the first switch on the network path between X and R. Consider a packet sent from X to Y over this connection.

Which of the following statements is/are TRUE about the destination IP and MAC addresses on this packet at the time it leaves X?

A. The destination IP address is the IP address of R
B. The destination IP address is the IP address of Y
C. The destination MAC address is the MAC address of S
D. The destination MAC address is the MAC address of Y

gatecse-2024-set2 computer-networks multiple-selects ethernet one-mark

Answer key

2.13.6 Ethernet: GATE CSE 2024 | Set 2 | Question: 45

Consider an Ethernet segment with a transmission speed of $10^8\,\mathrm{bits/sec}$ and a maximum segment length of 500 meters. If the speed of propagation of the signal in the medium is $2 \times 10^8\ \mathrm{meters/sec}$, then the minimum frame size (*in bits*) required for collision detection is ___________.

gatecse-2024-set2 numerical-answers computer-networks ethernet two-marks

Answer key

2.13.7 Ethernet: GATE IT 2006 | Question: 19

Which of the following statements is TRUE?

A. Both Ethernet frame and IP packet include checksum fields
B. Ethernet frame includes a checksum field and IP packet includes a CRC field
C. Ethernet frame includes a CRC field and IP packet includes a checksum field
D. Both Ethernet frame and IP packet include CRC fields

gateit-2006 computer-networks normal ethernet

Answer key

2.14 Fragmentation (5)

✍ **Practice Test:** Test 1 (12Q)

2.14.1 Fragmentation: GATE CSE 2004 | Question: 56

Consider three IP networks A, B and C. Host H_A in network A sends messages each containing 180 bytes of application data to a host H_C in network C. The TCP layer prefixes 20 byte header to the message.

This passes through an intermediate network B.The maximum packet size, including 20 byte IP header, in each network is:

- A: 1000 bytes
- B: 100 bytes
- C: 1000 bytes

The network A and B are connected through a 1 Mbps link, while B and C are connected by a 512 Kbps link (bps = bits per second).

Assuming that the packets are correctly delivered, how many bytes, including headers, are delivered to the IP layer at the destination for one application message, in the best case? Consider only data packets.

A. 200
B. 220
C. 240
D. 260

gatecse-2004 computer-networks ip-addressing fragmentation tcp normal

Answer key

2.14.2 Fragmentation: GATE CSE 2004 | Question: 57

Consider three IP networks A, B and C. Host H_A in network A sends messages each containing 180 $bytes$ of application data to a host H_C in network C. The TCP layer prefixes 20 byte header to the message. This passes through an intermediate network B. The maximum packet size, including 20 byte IP header, in each network, is:

- $A : 1000$ bytes
- $B : 100$ bytes
- $C : 1000$ bytes

The network A and B are connected through a 1 $Mbps$ link, while B and C are connected by a 512 $Kbps$ link (bps = bits per second).

What is the rate at which application data is transferred to host H_C? Ignore errors, acknowledgments, and other overheads.

A. 325.5 Kbps
B. 354.5 Kbps
C. 409.6 Kbps
D. 512.0 Kbps

gatecse-2004 computer-networks ip-addressing fragmentation tcp normal

Answer key

2.14.3 Fragmentation: GATE CSE 2013 | Question: 37

In an IPv4 datagram, the M bit is 0, the value of $HLEN$ is 10, the value of total length is 400 and the fragment offset value is 300. The position of the datagram, the sequence numbers of the first and the last bytes of the payload, respectively are:

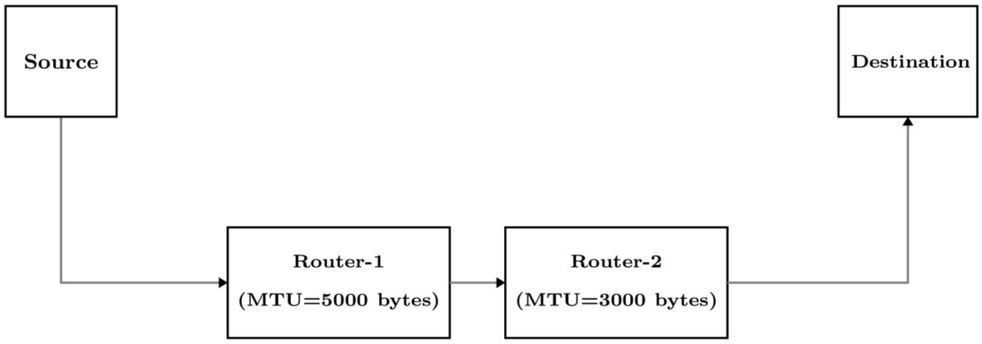
A. Last fragment, 2400 and 2789
B. First fragment, 2400 and 2759
C. Last fragment, 2400 and 2759
D. Middle fragment, 300 and 689

gatecse-2013 computer-networks ip-addressing fragmentation normal

Answer key

2.14.4 Fragmentation: GATE CSE 2025 | Set 1 | Question: 47

Suppose a message of size 15000 bytes is transmitted from a source to a destination using IPv4 protocol via two routers as shown in the figure. Each router has a defined maximum transmission unit (MTU) as shown in the figure, including IP header. The number of fragments that will be delivered to the destination is _________. (Answer in integer)



gatecse2025-set1 computer-networks ip-addressing fragmentation numerical-answers two-marks

Answer key

2.14.5 Fragmentation: GATE CSE 2025 | Set 2 | Question: 13

Consider a network that uses Ethernet and $IPv4$. Assume that $IPv4$ headers do not use any options field. Each Ethernet frame can carry a maximum of 1500 bytes in its data field. A UDP segment is transmitted. The payload (data) in the UDPO segment is 7488 bytes.

Which ONE of the following choices has the CORRECT total number of fragments transmitted and the size of the last fragment including $IPv4$ header?

A. 5 fragments 1488 bytes
B. 6 fragments 88 bytes
C. 6 fragments 108 bytes
D. 6 fragments 116 bytes

gatecse2025-set2 computer-networks fragmentation ip-addressing one-mark

Answer key

2.15 Hamming Code (2)

2.15.1 Hamming Code: GATE CSE 1994 | Question: 9

Following 7 bit single error correcting hamming coded message is received.

7	6	5	4	3	2	1	bit No.
1	0	0	0	1	1	0	X

Determine if the message is correct (assuming that at most 1 bit could be corrupted). If the message contains an error find the bit which is erroneous and gives correct message.

gate1994 computer-networks error-detection hamming-code normal descriptive

Answer key

2.15.2 Hamming Code: GATE CSE 2021 | Set 1 | Question: 29

Assume that a 12-bit Hamming codeword consisting of 8-bit data and 4 check bits is $d_8d_7d_6d_5c_8d_4d_4d_3d_2c_4d_1c_2c_1$, where the data bits and the check bits are given in the following tables:

Data bits

d_8	d_7	d_6	d_5	d_4	d_3	d_2	d_1
1	1	0	x	0	1	0	1

Check bits

c_8	c_4	c_2	c_1
y	0	1	0

Which one of the following choices gives the correct values of x and y?

A. x is 0 and y is 0 B. x is 0 and y is 1 C. x is 1 and y is 0 D. x is 1 and y is 1

gatecse-2021-set1 computer-networks hamming-code two-marks error-detection

Answer key

2.16 IP Addressing (8)

Practice Tests: Test 1 (15Q) Test 2 (2Q)

2.16.1 IP Addressing: GATE CSE 2003 | Question: 27

Which of the following assertions is FALSE about the Internet Protocol (IP)?

A. It is possible for a computer to have multiple IP addresses
B. IP packets from the same source to the same destination can take different routes in the network
C. IP ensures that a packet is discarded if it is unable to reach its destination within a given number of hops
D. The packet source cannot set the route of an outgoing packets; the route is determined only by the routing tables in the routers on the way

gatecse-2003 computer-networks ip-addressing normal

Answer key

2.16.2 IP Addressing: GATE CSE 2012 | Question: 23

In the IPv4 addressing format, the number of networks allowed under Class C addresses is:

A. 2^{14} B. 2^{7} C. 2^{21} D. 2^{24}

gatecse-2012 computer-networks ip-addressing easy

Answer key

2.16.3 IP Addressing: GATE CSE 2017 | Set 2 | Question: 20

The maximum number of **IPv4** router addresses that can be listed in the record route (RR) option field of an **IPv4** header is______.

gatecse-2017-set2 computer-networks ip-addressing numerical-answers

Answer key

2.16.4 IP Addressing: GATE CSE 2018 | Question: 54

Consider an IP packet with a length of $4,500$ bytes that includes a 20-byte **IPv4** header and a 40-byte TCP header. The packet is forwarded to an **IPv4** router that supports a Maximum Transmission Unit (MTU) of 600 bytes. Assume that the length of the IP header in all the outgoing fragments of this packet is 20 bytes. Assume that the fragmentation offset value stored in the first fragment is 0.

The fragmentation offset value stored in the third fragment is ________.

gatecse-2018 computer-networks ip-addressing numerical-answers two-marks

Answer key

2.16.5 IP Addressing: GATE CSE 2023 | Question: 42

Suppose in a web browser, you click on the **www. gate-2023.in** URL. The browser cache is empty. The IP address for this URL is not cached in your local host, so a DNS lookup is triggered (by the local DNS server deployed on your local host) over the 3-tier DNS hierarchy in an iterative mode. No resource records are cached anywhere across all DNS servers.

Let RTT denote the round trip time between your local host and DNS servers in the DNS hierarchy. The round trip time between the local host and the web server hosting **www.gate-2023.in** is also equal to RTT. The HTML file associated with the URL is small enough to have negligible transmission time and negligible rendering time by your web browser, which references 10 equally small objects on the same web server.

Which of the following statements is/are CORRECT about the minimum elapsed time between clicking on the URL and your browser fully rendering it?

A. 7 RTTs, in case of non-persistent HTTP with 5 parallel TCP connections.
B. 5 RTTs, in case of persistent HTTP with pipelining.
C. 9 RTTs, in case of non-persistent HTTP with 5 parallel TCP connections.
D. 6 RTTs, in case of persistent HTTP with pipelining.

gatecse-2023 computer-networks ip-addressing multiple-selects two-marks

Answer key

2.16.6 IP Addressing: GATE CSE 2024 | Set 2 | Question: 28

Which one of the following CIDR prefixes exactly represents the range of IP addresses $10.12.2.0$ to $10.12.3.255$?

A. $10.12.2.0/23$
B. $10.12.2.0/24$
C. $10.12.0.0/22$
D. $10.12.2.0/22$

gatecse-2024-set2 computer-networks ip-addressing two-marks

Answer key

2.16.7 IP Addressing: GATE CSE 2025 | Set 2 | Question: 8

A machine receives an $IPv4$ datagram. The protocol field of the $IPv4$ header has the protocol number of a protocol X.

Which ONE of the following is NOT a possible candidate for X?

A. Internet Control Message Protocol (ICMP)
B. Internet Group Management (IGMP)
C. Open Shortest Path First (OSPF)
D. Routing Information Protocol (RIP)

gatecse2025-set2 computer-networks ip-addressing one-mark

Answer key

2.16.8 IP Addressing: GATE CSE 2026 | Set 2 | Question: 23

If an IP network uses a subnet mask of $255.255.240.0$, the maximum number of IP addresses that can be assigned to network interfaces is _____. (answer in integer)

gatecse-2026-set2 computer-networks ip-addressing numerical-answers one-mark

Answer key

2.17 IP Packet (12)

✍ **Practice Tests:** Test 1 (15Q) Test 2 (3Q)

2.17.1 IP Packet: GATE CSE 2006 | Question: 5

For which one of the following reasons does internet protocol(IP) use the time-to-live(TTL) field in IP datagram header?

A. Ensure packets reach destination within that time
B. Discard packets that reach later than that time
C. Prevent packets from looping indefinitely
D. Limit the time for which a packet gets queued in intermediate routers

gatecse-2006 computer-networks ip-addressing ip-packet easy

Answer key

2.17.2 IP Packet: GATE CSE 2010 | Question: 15. PGEE 2018

One of the header fields in an IP datagram is the Time-to-Live (TTL) field. Which of the following statements best explains the need for this field?

A. It can be used to prioritize packets.
B. It can be used to reduce delays.
C. It can be used to optimize throughput.
D. It can be used to prevent packet looping.

gatecse-2010 computer-networks ip-packet easy

Answer key

2.17.3 IP Packet: GATE CSE 2014 | Set 3 | Question: 25

Host A (on TCP/IP v4 network A) sends an IP datagram D to host B (also on TCP/IP v4 network B). Assume that no error occurred during the transmission of D. When D reaches B, which of the following IP header field(s) may be different from that of the original datagram D?

i. TTL
ii. Checksum
iii. Fragment Offset

A. i only
B. i and ii only
C. ii and iii only
D. i, ii and iii

gatecse-2014-set3 computer-networks ip-packet normal

Answer key

2.17.4 IP Packet: GATE CSE 2014 | Set 3 | Question: 28

An IP router with a $\text{Maximum Transmission Unit (MTU)}$ of 1500 bytes has received an IP packet of size 4404 bytes with an IP header of length 20 bytes. The values of the relevant fields in the header of the third IP fragment generated by the router for this packet are:

A. $\text{MF bit}: 0,$ Datagram Length:$1444;$ Offset: 370
B. $\text{MF bit}: 1,$ Datagram Length: $1424;$ Offset: 185
C. $\text{MF bit}: 1,$ Datagram Length: $1500;$ Offset: 370
D. $\text{MF bit}: 0,$ Datagram Length: $1424;$ Offset: 2960

gatecse-2014-set3 computer-networks fragmentation ip-packet normal

Answer key

2.17.5 IP Packet: GATE CSE 2015 | Set 1 | Question: 22

Which of the following fields of an IP header is NOT modified by a typical IP router?

A. Check sum
B. Source address
C. Time to Live (TTL)
D. Length

gatecse-2015-set1 computer-networks ip-packet easy

Answer key

2.17.6 IP Packet: GATE CSE 2015 | Set 2 | Question: 52

Host A sends a UDP datagram containing 8880 bytes of user data to host B over an $\text{Ethernet LAN}.$ Ethernet frames may carry data up to $1500 \text{ bytes (i.e. MTU} = 1500 \text{ bytes)}.$ Size of UDP header is 8 bytes and size of IP header is $20 \text{ bytes}.$ There is no option field in IP header. How many total number of IP fragments will be transmitted and what will be the contents of offset field in the last fragment?

A. 6 and 925
B. 6 and 7400
C. 7 and 1110
D. 7 and 8880

gatecse-2015-set2 computer-networks fragmentation ip-packet normal

Answer key

2.17.7 IP Packet: GATE CSE 2016 | Set 1 | Question: 53

An IP datagram of size 1000 bytes arrives at a router. The router has to forward this packet on a link whose MTU (maximum transmission unit) is 100 bytes . Assume that the size of the IP header is 20 bytes .

The number of fragments that the IP datagram will be divided into for transmission is________.

gatecse-2016-set1 computer-networks fragmentation ip-packet normal numerical-answers

Answer key

2.17.8 IP Packet: GATE CSE 2024 | Set 1 | Question: 21

Which of the following fields is/are modified in the IP header of a packet going out of a network address translation (NAT) device from an internal network to an external network?

A. Source IP
B. Destination IP
C. Header Checksum
D. Total Length

gatecse-2024-set1 multiple-selects computer-networks ip-packet one-mark

Answer key

2.17.9 IP Packet: GATE CSE 2024 | Set 1 | Question: 55

Consider sending an IP datagram of size 1420 bytes (including 20 bytes of IP header) from a sender to a receiver over a path of two links with a router between them. The first link (sender to router) has an MTU (Maximum Transmission Unit) size of 542 bytes, while the second link (router to receiver) has an MTU size of 360 bytes. The number of fragments that would be delivered at the receiver is ____________.

gatecse-2024-set1 numerical-answers computer-networks ip-packet fragmentation two-marks

Answer key

2.17.10 IP Packet: GATE CSE 2024 | Set 2 | Question: 18

Which of the following statements about IPv4 fragmentation is/are TRUE?

A. The fragmentation of an IP datagram is performed *only* at the source of the datagram
B. The fragmentation of an IP datagram is performed at any IP router which finds that the size of the datagram to be transmitted exceeds the MTU
C. The reassembly of fragments is performed *only* at the destination of the datagram
D. The reassembly of fragments is performed at all intermediate routers along the path from the source to the destination

gatecse-2024-set2 computer-networks multiple-selects ip-packet one-mark

Answer key

2.17.11 IP Packet: GATE CSE 2024 | Set 2 | Question: 22

Which of the following fields of an IP header is/are *always* modified by any router before it forwards the IP packet?

A. Source IP Address
B. Protocol
C. Time to Live (TTL)
D. Header Checksum

gatecse-2024-set2 computer-networks multiple-selects ip-packet easy one-mark

Answer key

2.17.12 IP Packet: GATE IT 2004 | Question: 86

In the TCP/IP protocol suite, which one of the following is NOT part of the IP header?

A. Fragment Offset
B. Source IP address
C. Destination IP address
D. Destination port number

gateit-2004 computer-networks ip-packet easy

Answer key

2.18 Icmp (1)

2.18.1 Icmp: GATE IT 2005 | Question: 26

Traceroute reports a possible route that is taken by packets moving from some host A to some other host B. Which of the following options represents the technique used by traceroute to identify these hosts:

A. By progressively querying routers about the next router on the path to B using ICMP packets, starting with the first router
B. By requiring each router to append the address to the ICMP packet as it is forwarded to B. The list of all routers en-route to B is returned by B in an ICMP reply packet
C. By ensuring that an ICMP reply packet is returned to A by each router en-route to B, in the ascending order of their hop distance from A
D. By locally computing the shortest path from A to B

gateit-2005 computer-networks icmp application-layer-protocols normal

Answer key

2.19 LAN Technologies (7)

Practice Test: Test 1 (11Q)

2.19.1 LAN Technologies: GATE CSE 2003 | Question: 83

A $2\ km$ long broadcast LAN has 10^7 bps bandwidth and uses CSMA/CD. The signal travels along the wire at 2×10^8 m/s. What is the minimum packet size that can be used on this network?

A. 50 bytes B. 100 bytes C. 200 bytes D. None of the above

gatecse-2003 computer-networks lan-technologies normal

Answer key

2.19.2 LAN Technologies: GATE CSE 2007 | Question: 65

There are n stations in slotted LAN. Each station attempts to transmit with a probability p in each time slot. What is the probability that **ONLY** one station transmits in a given time slot?

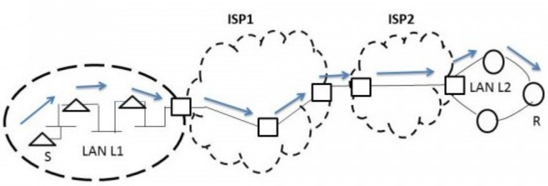
A. $np(1-p)^{n-1}$ B. $(1-p)^{n-1}$ C. $p(1-p)^{n-1}$ D. $1-(1-p)^{n-1}$

gatecse-2007 computer-networks lan-technologies probability normal

Answer key

2.19.3 LAN Technologies: GATE CSE 2014 | Set 2 | Question: 25

In the diagram shown below, $L1$ is an Ethernet LAN and $L2$ is a Token-Ring LAN. An IP packet originates from sender S and traverses to R, as shown. The links within each ISP and across the two ISPs, are all point-to-point optical links. The initial value of the TTL field is 32. The maximum possible value of the TTL field when R receives the datagram is _______.



gatecse-2014-set2 computer-networks numerical-answers lan-technologies ethernet normal

Answer key

2.19.4 LAN Technologies: GATE CSE 2019 | Question: 49

Consider that 15 machines need to be connected in a LAN using 8-port Ethernet switches. Assume that these switches do not have any separate uplink ports. The minimum number of switches needed is _______

gatecse-2019 numerical-answers computer-networks lan-technologies two-marks

Answer key☟

2.19.5 LAN Technologies: GATE IT 2004 | Question: 27

A host is connected to a Department network which is part of a University network. The University network, in turn, is part of the Internet. The largest network in which the Ethernet address of the host is unique is

A. the subnet to which the host belongs
B. the Department network
C. the University network
D. the Internet

gateit-2004 computer-networks lan-technologies ethernet normal

Answer key☟

2.19.6 LAN Technologies: GATE IT 2005 | Question: 28

Which of the following statements is FALSE regarding a bridge?

A. Bridge is a layer 2 device
B. Bridge reduces collision domain
C. Bridge is used to connect two or more LAN segments
D. Bridge reduces broadcast domain

gateit-2005 computer-networks lan-technologies normal

Answer key☟

2.19.7 LAN Technologies: GATE IT 2006 | Question: 66

A router has two full-duplex Ethernet interfaces each operating at 100 Mb/s. Ethernet frames are at least 84 bytes long (including the Preamble and the Inter-Packet-Gap). The maximum packet processing time at the router for wirespeed forwarding to be possible is (in microseconds)

A. 0.01
B. 3.36
C. 6.72
D. 8

gateit-2006 computer-networks lan-technologies ethernet normal

Answer key☟

2.20 MAC Protocol (4)

✍ **Practice Test:** Test 1 (6Q)

2.20.1 MAC Protocol: GATE CSE 2005 | Question: 74

Suppose the round trip propagation delay for a 10 Mbps Ethernet having 48-bit jamming signal is $46.4 \ \mu s$. The minimum frame size is:

A. 94
B. 416
C. 464
D. 512

gatecse-2005 computer-networks mac-protocol ethernet

Answer key☟

2.20.2 MAC Protocol: GATE CSE 2015 | Set 2 | Question: 8

A link has transmission speed of 10^6 bits/sec. It uses data packets of size 1000 bytes each. Assume that the acknowledgment has negligible transmission delay and that its propagation delay is the same as the data propagation delay. Also, assume that the processing delays at nodes are negligible. The efficiency of the stop-and-wait protocol in this setup is exactly 25%. The value of the one way propagation delay (in milliseconds) is_____.

gatecse-2015-set2 computer-networks mac-protocol stop-and-wait normal numerical-answers

Answer key☟

2.20.3 MAC Protocol: GATE IT 2004 | Question: 85

Consider a simplified time slotted MAC protocol, where each host always has data to send and transmits with probability $p = 0.2$ in every slot. There is no backoff and one frame can be transmitted in one slot. If more than one host transmits in the same slot, then the transmissions are unsuccessful due to collision. What is the maximum number of hosts which this protocol can support if each host has to be provided a minimum throughput of 0.16 frames per time slot?

A. 1 B. 2 C. 3 D. 4

gateit-2004 computer-networks congestion-control mac-protocol normal

Answer key☟

2.20.4 MAC Protocol: GATE IT 2005 | Question: 75

In a TDM medium access control bus LAN, each station is assigned one time slot per cycle for transmission. Assume that the length of each time slot is the time to transmit 100 bits plus the end-to-end propagation delay. Assume a propagation speed of $2 \times 10^8 m/sec$. The length of the LAN is 1 km with a bandwidth of 10 Mbps. The maximum number of stations that can be allowed in the LAN so that the throughput of each station can be $2/3$ Mbps is

A. 3 B. 5 C. 10 D. 20

gateit-2005 computer-networks mac-protocol normal

Answer key☟

2.21 Network Flow (4)

✍ **Practice Test:** Test 1 (5Q)

2.21.1 Network Flow: GATE CSE 1992 | Question: 01,v

A simple and reliable data transfer can be accomplished by using the 'handshake protocol'. It accomplishes reliable data transfer because for every data item sent by the transmitter ________.

gate1992 computer-networks network-flow easy fill-in-the-blanks

Answer key☟

2.21.2 Network Flow: GATE CSE 2017 | Set 2 | Question: 35

Consider two hosts X and Y, connected by a single direct link of rate 10^6 bits/sec. The distance between the two hosts is $10,000$ km and the propagation speed along the link is 2×10^8 m/sec. Host X sends a file of $50,000$ bytes as one large message to host Y continuously. Let the transmission and propagation delays be p milliseconds and q milliseconds respectively. Then the value of p and q are

A. $p = 50$ and $q = 100$ B. $p = 50$ and $q = 400$ C. $p = 100$ and $q = 50$ D. $p = 400$ and $q = 50$

gatecse-2017-set2 computer-networks network-flow

Answer key☟

2.21.3 Network Flow: GATE IT 2004 | Question: 87

A TCP message consisting of 2100 *bytes* is passed to IP for delivery across two networks. The first network can carry a maximum payload of 1200 *bytes* per frame and the second network can carry a maximum payload of 400 *bytes* per frame, excluding network overhead. Assume that IP overhead per packet is 20 *bytes*. What is the total IP overhead in the second network for this transmission?

A. 40 bytes B. 80 bytes C. 120 bytes D. 160 bytes

gateit-2004 computer-networks network-flow normal

Answer key☟

2.21.4 Network Flow: GATE IT 2006 | Question: 67

A link of capacity 100 Mbps is carrying traffic from a number of sources. Each source generates an on-off traffic stream; when the source is on, the rate of traffic is 10 Mbps, and when the source is off, the rate of traffic is zero. The duty cycle, which is the ratio of on-time to off-time, is $1 : 2$. When there is no buffer at the link, the minimum number of sources that can be multiplexed on the link so that link capacity is not wasted and no data loss occurs is $S1$. Assuming that all sources are synchronized and that the link is provided with a large buffer, the maximum number of sources that can be multiplexed so that no data loss occurs is $S2$. The values of $S1$ and $S2$ are, respectively,

A. 10 and 30
B. 12 and 25
C. 5 and 33
D. 15 and 22

gateit-2006 computer-networks network-flow normal

Answer key

2.22 Network Layer (6)

2.22.1 Network Layer: GATE CSE 2003 | Question: 28

Which of the following functionality *must* be implemented by a transport protocol over and above the network protocol?

A. Recovery from packet losses
B. Detection of duplicate packets
C. Packet delivery in the correct order
D. End to end connectivity

gatecse-2003 computer-networks network-layer easy

Answer key

2.22.2 Network Layer: GATE CSE 2004 | Question: 15

Choose the best matching between Group 1 and Group 2

Group-1	**Group-2**
P. Data link layer	1. Ensures reliable transport of data over a physical point-to-point link
Q. Network layer	2. Encodes/decodes data for physical transmission
R. Transport layer	3. Allows end-to-end communication between two processes
	4. Routes data from one network node to the next

A. P-1, Q-4, R-3
B. P-2, Q-4, R-1
C. P-2, Q-3, R-1
D. P-1, Q-3, R-2

gatecse-2004 computer-networks network-layer normal

Answer key

2.22.3 Network Layer: GATE CSE 2013 | Question: 14

Assume that source S and destination D are connected through two intermediate routers labeled R. Determine how many times each packet has to visit the network layer and the data link layer during a transmission from S to D.

A. Network layer – 4 times and Data link layer – 4 times
B. Network layer – 4 times and Data link layer – 3 times
C. Network layer – 4 times and Data link layer – 6 times
D. Network layer – 2 times and Data link layer – 6 times

gatecse-2013 computer-networks network-layer normal

Answer key

2.22.4 Network Layer: GATE CSE 2014 | Set 3 | Question: 23

In the following pairs of OSI protocol layer/sub-layer and its functionality, the **INCORRECT** pair is

A. Network layer and Routing
B. Data Link Layer and Bit synchronization
C. Transport layer and End-to-end process communication
D. Medium Access Control sub-layer and Channel sharing

gatecse-2014-set3 computer-networks network-layer easy

Answer key

2.22.5 Network Layer: GATE CSE 2018 | Question: 13

Match the following:

Field	**Length in bits**
P. UDP Header's Port Number	I. 48
Q. Ethernet MAC Address	II. 8
R. IPv6 Next Header	III. 32
S. TCP Header's Sequence Number	IV. 16

A. P-III, Q-IV, R-II, S-I
B. P-II, Q-I, R-IV, S-III
C. P-IV, Q-I, R-II, S-III
D. P-IV, Q-I, R-III, S-II

gatecse-2018 computer-networks network-layer match-the-following easy one-mark

Answer key

2.22.6 Network Layer: GATE CSE 2026 | Set 1 | Question: 46

An ISP having an address block $202.16.0.0/15$ assigns a block of 6000 IP addresses to a client, using the classless internet domain routing (CIDR) super-netting approach. Which of the following address blocks can be assigned by the ISP?

A. $202.16.0.0/19$
B. $202.17.64.0/19$
C. $202.16.32.0/19$
D. $202.17.24.0/19$

gatecse-2026-set1 computer-networks two-marks network-layer multiple-selects

Answer key

2.23 Network Protocols (11)

✍ **Practice Test:** Test 1 (15Q)

2.23.1 Network Protocols: GATE CSE 2005 | Question: 24

The address resolution protocol (ARP) is used for:

A. Finding the IP address from the DNS
B. Finding the IP address of the default gateway
C. Finding the IP address that corresponds to a MAC address
D. Finding the MAC address that corresponds to an IP address

gatecse-2005 computer-networks normal network-protocols

Answer key

2.23.2 Network Protocols: GATE CSE 2007 | Question: 20

Which one of the following uses UDP as the transport protocol?

A. HTTP
B. Telnet
C. DNS
D. SMTP

gatecse-2007 computer-networks network-protocols application-layer-protocols easy

Answer key

2.23.3 Network Protocols: GATE CSE 2007 | Question: 70

Match the following:

(P)	SMTP	(1)	Application layer
(Q)	BGP	(2)	Transport layer
(R)	TCP	(3)	Data link layer
(S)	PPP	(4)	Network layer
		(5)	Physical layer

A. P - 2, Q - 1, R - 3, S - 5
B. P - 1, Q - 4, R - 2, S - 3
C. P - 1, Q - 4, R - 2, S - 5
D. P - 2, Q - 4, R - 1, S - 3

gatecse-2007 computer-networks network-layer network-protocols easy match-the-following

Answer key☟

2.23.4 Network Protocols: GATE CSE 2015 | Set 1 | Question: 17

In one of the pairs of protocols given below , both the protocols can use multiple TCP connections between the same client and the server. Which one is that?

A. HTTP, FTP
B. HTTP, TELNET
C. FTP, SMTP
D. HTTP, SMTP

gatecse-2015-set1 computer-networks network-protocols normal

Answer key☟

2.23.5 Network Protocols: GATE CSE 2016 | Set 1 | Question: 24

Which one of the following protocols is **NOT** used to resolve one form of address to another one?

A. DNS
B. ARP
C. DHCP
D. RARP

gatecse-2016-set1 computer-networks network-protocols easy

Answer key☟

2.23.6 Network Protocols: GATE CSE 2019 | Question: 29

Suppose that in an IP-over-Ethernet network, a machine X wishes to find the MAC address of another machine Y in its subnet. Which one of the following techniques can be used for this?

A. X sends an ARP request packet to the local gateway's IP address which then finds the MAC address of Y and sends to X
B. X sends an ARP request packet to the local gateway's MAC address which then finds the MAC address of Y and sends to X
C. X sends an ARP request packet with broadcast MAC address in its local subnet
D. X sends an ARP request packet with broadcast IP address in its local subnet

gatecse-2019 computer-networks network-protocols two-marks

Answer key☟

2.23.7 Network Protocols: GATE CSE 2021 | Set 1 | Question: 49

Consider the sliding window flow-control protocol operating between a sender and a receiver over a full-duplex error-free link. Assume the following:

- The time taken for processing the data frame by the receiver is negligible.
- The time taken for processing the acknowledgement frame by the sender is negligible.
- The sender has infinite number of frames available for transmission.
- The size of the data frame is $2,000$ bits and the size of the acknowledgement frame is 10 bits.
- The link data rate in each direction is 1 Mbps ($= 10^6$ bits per second).
- One way propagation delay of the link is 100 milliseconds.

The minimum value of the sender's window size in terms of the number of frames, (rounded to the nearest integer) needed to achieve a link utilization of 50% is__________.

gatecse-2021-set1 computer-networks network-protocols sliding-window numerical-answers two-marks

Answer key

2.23.8 Network Protocols: GATE CSE 2021 | Set 1 | Question: 8

Consider the following two statements.

- S_1: Destination MAC address of an ARP reply is a broadcast address.
- S_2: Destination MAC address of an ARP request is a broadcast address.

Which one of the following choices is correct?

A. Both S_1 and S_2 are true
B. S_1 is true and S_2 is false
C. S_1 is false and S_2 is true
D. Both S_1 and S_2 are false

gatecse-2021-set1 computer-networks network-protocols one-mark

Answer key

2.23.9 Network Protocols: GATE CSE 2024 | Set 1 | Question: 6

A user starts browsing a webpage hosted at a remote server. The browser opens a single TCP connection to fetch the entire webpage from the server. The webpage consists of a top-level index page with multiple embedded image objects. Assume that all caches (e.g., DNS cache, browser cache) are all initially empty. The following packets leave the user's computer in some order.

i. HTTP GET request for the index page
ii. DNS request to resolve the web server's name to its IP address
iii. HTTP GET request for an image object
iv. TCP SYN to open a connection to the web server

Which one of the following is the CORRECT chronological order (earliest in time to latest) of the packets leaving the computer?

A. $\text{(iv), (ii), (iii), (i)}$
B. $\text{(ii), (iv), (iii), (i)}$
C. $\text{(ii), (iv), (i), (iii)}$
D. $\text{(iv), (ii), (i), (iii)}$

gatecse-2024-set1 computer-networks network-protocols one-mark

Answer key

2.23.10 Network Protocols: GATE IT 2007 | Question: 69

Consider the following clauses:

i. Not inherently suitable for client authentication.
ii. Not a state sensitive protocol.
iii. Must be operated with more than one server.
iv. Suitable for structured message organization.
v. May need two ports on the serve side for proper operation.

The option that has the maximum number of correct matches is

A. IMAP-i; FTP-ii; HTTP-iii; DNS-iv; POP3-v
B. FTP-i; POP3-ii; SMTP-iii; HTTP-iv; IMAP-v
C. POP3-i; SMTP-ii; DNS-iii; IMAP-iv; HTTP-v
D. SMTP-i; HTTP-ii; IMAP-iii; DNS-iv; FTP-v

gateit-2007 computer-networks network-protocols normal

Answer key

2.23.11 Network Protocols: GATE IT 2008 | Question: 68

Which of the following statements are TRUE?

- **S1:** TCP handles both congestion and flow control
- **S2:** UDP handles congestion but not flow control
- **S3:** Fast retransmit deals with congestion but not flow control
- **S4:** Slow start mechanism deals with both congestion and flow control

A. $S1$, $S2$ and $S3$ only
B. $S1$ and $S3$ only
C. $S3$ and $S4$ only
D. $S1$, $S3$ and $S4$ only

gateit-2008 computer-networks network-protocols normal

Answer key

2.24 Network Switching (4)

Practice Test: Test 1 (6Q)

2.24.1 Network Switching: GATE CSE 2005 | Question: 73

In a packet switching network, packets are routed from source to destination along a single path having two intermediate nodes. If the message size is 24 bytes and each packet contains a header of 3 bytes, then the optimum packet size is:

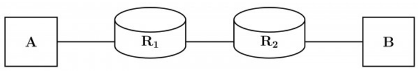
A. 4
B. 6
C. 7
D. 9

gatecse-2005 computer-networks network-switching normal

Answer key

2.24.2 Network Switching: GATE CSE 2014 | Set 2 | Question: 26

Consider the store and forward packet switched network given below. Assume that the bandwidth of each link is 10^6 bytes / sec. A user on host A sends a file of size 10^3 bytes to host B through routers $R1$ and $R2$ in three different ways. In the first case a single packet containing the complete file is transmitted from A to B. In the second case, the file is split into 10 equal parts, and these packets are transmitted from A to B. In the third case, the file is split into 20 equal parts and these packets are sent from A to B. Each packet contains 100 bytes of header information along with the user data. Consider only transmission time and ignore processing, queuing and propagation delays. Also assume that there are no errors during transmission. Let $T1$, $T2$ and $T3$ be the times taken to transmit the file in the first, second and third case respectively. Which one of the following is CORRECT?



A. $T1 < T2 < T3$
B. $T1 > T2 > T3$
C. $T2 = T3, T3 < T1$
D. $T1 = T3, T3 > T2$

gatecse-2014-set2 computer-networks network-switching normal

Answer key

2.24.3 Network Switching: GATE CSE 2015 | Set 3 | Question: 36

Two hosts are connected via a packet switch with 10^7 bits per second links. Each link has a propagation delay of 20 microseconds. The switch begins forwarding a packet 35 microseconds after it receives the same. If 10000 bits of data are to be transmitted between the two hosts using a packet size of 5000 bits, the time elapsed between the transmission of the first bit of data and the reception of the last bit of the data in microseconds is ______.

gatecse-2015-set3 computer-networks normal numerical-answers network-switching

Answer key

2.24.4 Network Switching: GATE IT 2004 | Question: 22

Which one of the following statements is FALSE?

A. Packet switching leads to better utilization of bandwidth resources than circuit switching

B. Packet switching results in less variation in delay than circuit switching
C. Packet switching requires more per-packet processing than circuit switching
D. Packet switching can lead to reordering unlike in circuit switching

gateit-2004 computer-networks network-switching normal

Answer key

2.25 Osi Model (1)

2.25.1 Osi Model: GATE CSE 2025 | Set 1 | Question: 6

Identify the ONE CORRECT matching between the OSI layers and their corresponding functionalities as shown.

	OSI Layers		**Functionalities**
(a)	Network Layer	(I)	Packet Routing
(b)	Transport Layer	(II)	Framing and error handling
(c)	Datalink layer	(III)	Host to host communication

A. (a)-(I), (b)-(II), (c)-(III)
B. (a)-(I), (b)-(III), (c)-(II)
C. (a)-(II), (b)-(I), (c)-(III)
D. (a)-(III), (b)-(II), (c)-(I)

gatecse2025-set1 computer-networks osi-model easy one-mark

Answer key

2.26 Probability (1)

2.26.1 Probability: GATE CSE 2025 | Set 1 | Question: 46

Suppose a 5-bit message is transmitted from a source to a destination through a noisy channel. The probability that a bit of the message gets flipped during transmission is 0.01. Flipping of each bit is independent of one another. The probability that the message is delivered error-free to the destination is ___________ (rounded off to three decimal places)

gatecse2025-set1 computer-networks probability numerical-answers easy two-marks

Answer key

2.27 Pure Aloha (1)

2.27.1 Pure Aloha: GATE CSE 2021 | Set 2 | Question: 54

Consider a network using the pure ALOHA medium access control protocol, where each frame is of length $1,000$ bits. The channel transmission rate is 1 Mbps $(= 10^6$ bits per second). The aggregate number of transmissions across all the nodes (including new frame transmissions and retransmitted frames due to collisions) is modelled as a Poisson process with a rate of $1,000$ frames per second. Throughput is defined as the average number of frames successfully transmitted per second. The throughput of the network (rounded to the nearest integer) is ______________

gatecse-2021-set2 computer-networks mac-protocol pure-aloha numerical-answers two-marks

Answer key

2.28 Routing (14)

✍ **Practice Tests:** Test 1 (15Q) Test 2 (4Q)

2.28.1 Routing: GATE CSE 2005 | Question: 26

In a network of LANs connected by bridges, packets are sent from one LAN to another through intermediate bridges. Since more than one path may exist between two LANs, packets may have to be routed through multiple bridges. Why is the *spanning tree algorithm* used for bridge-routing?

A. For shortest path routing between LANs
B. For avoiding loops in the routing paths

C. For fault tolerance

D. For minimizing collisions

gatecse-2005 computer-networks routing normal

Answer key

2.28.2 Routing: GATE CSE 2014 | Set 1 | Question: 23

Consider the following three statements about link state and distance vector routing protocols, for a large network with 500 network nodes and 4000 links.

[S1]: The computational overhead in link state protocols is higher than in distance vector protocols.

[S2]: A distance vector protocol (with split horizon) avoids persistent routing loops, but not a link state protocol.

[S3]: After a topology change, a link state protocol will converge faster than a distance vector protocol.

Which one of the following is correct about $S1$, $S2$, and $S3$?

A. $S1$, $S2$, and $S3$ are all true.

B. $S1$, $S2$, and $S3$ are all false.

C. $S1$ and $S2$ are true, but $S3$ is false.

D. $S1$ and $S3$ are true, but $S2$ is false.

gatecse-2014-set1 computer-networks routing normal

Answer key

2.28.3 Routing: GATE CSE 2014 | Set 2 | Question: 23

Which of the following is TRUE about the interior gateway routing protocols — Routing Information Protocol (RIP) and Open Shortest Path First $(OSPF)$

A. RIP uses distance vector routing and OSPF uses link state routing
B. OSPF uses distance vector routing and RIP uses link state routing
C. Both RIP and OSPF use link state routing
D. Both RIP and OSPF use distance vector routing

gatecse-2014-set2 computer-networks routing easy

Answer key

2.28.4 Routing: GATE CSE 2014 | Set 3 | Question: 26

An IP router implementing Classless Inter-domain Routing (CIDR) receives a packet with address $131.23.151.76$. The router's routing table has the following entries:

Prefix	Outer Interface Identifier
$131.16.0.0/12$	3
$131.28.0.0/14$	5
$131.19.0.0/16$	2
$131.22.0.0/15$	1

The identifier of the output interface on which this packet will be forwarded is ______.

gatecse-2014-set3 computer-networks routing normal numerical-answers

Answer key

2.28.5 Routing: GATE CSE 2017 | Set 2 | Question: 09

Consider the following statements about the routing protocols. Routing Information Protocol (RIP) and Open Shortest Path First (OSPF) in an IPv4 network.

I. RIP uses distance vector routing
II. RIP packets are sent using UDP
III. OSPF packets are sent using TCP
IV. OSPF operation is based on link-state routing

Which of the above statements are CORRECT?

A. I and IV only
B. I, II and III only
C. I, II and IV only
D. II, III and IV only

gatecse-2017-set2 computer-networks routing

Answer key☟

2.28.6 Routing: GATE CSE 2020 | Question: 15

Consider the following statements about the functionality of an IP based router.

I. A router does not modify the IP packets during forwarding.
II. It is not necessary for a router to implement any routing protocol.
III. A router should reassemble IP fragments if the MTU of the outgoing link is larger than the size of the incoming IP packet.

Which of the above statements is/are TRUE?

A. I and II only
B. I only
C. II and III only
D. II only

gatecse-2020 computer-networks routing one-mark

Answer key☟

2.28.7 Routing: GATE CSE 2023 | Question: 15

Which of the following statements is/are INCORRECT about the OSPF (Open Shortest Path First) routing protocol used in the Internet?

A. OSPF implements Bellman-Ford algorithm to find shortest paths.
B. OSPF uses Dijkstra's shortest path algorithm to implement least-cost path routing.
C. OSPF is used as an inter-domain routing protocol.
D. OSPF implements hierarchical routing.

gatecse-2023 computer-networks routing multiple-selects one-mark

Answer key☟

2.28.8 Routing: GATE CSE 2024 | Set 1 | Question: 26

Consider a network path $\text{P} - \text{Q} - \text{R}$ between nodes P and R via router Q. Node P sends a file of size 10^6 bytes to R via this path by splitting the file into chunks of 10^3 bytes each. Node P sends these chunks one after the other without any wait time between the successive chunk transmissions. Assume that the size of extra headers added to these chunks is negligible, and that the chunk size is less than the MTU.

Each of the links $\text{P} - \text{Q}$ and $\text{Q} - \text{R}$ has a bandwidth of 10^6bits/sec, and negligible propagation latency. Router Q immediately transmits every packet it receives from P to R, with negligible processing and queueing delays. Router Q can simultaneously receive on $\text{link}\, \text{P} - \text{Q}$ and transmit on $\text{link}\, \text{Q} - \text{R}$.

Assume P starts transmitting the chunks at time $t = 0$.

Which one of the following options gives the time (*in seconds, rounded off to* 3 *decimal places*) at which R receives all the chunks of the file?

A. 8.000
B. 8.008
C. 15.992
D. 16.000

gatecse-2024-set1 computer-networks routing two-marks

Answer key☟

2.28.9 Routing: GATE CSE 2024 | Set 1 | Question: 48

Consider the entries shown below in the forwarding table of an IP router. Each entry consists of an IP prefix and the corresponding next hop router for packets whose destination IP address matches the prefix. The notation " /N " in a prefix indicates a subnet mask with the most significant N bits set to 1.

Prefix	Next hop router
10.1.1.0 / 24	R1
10.1.1.128 / 25	R2
10.1.1.64 / 26	R3
10.1.1.192 / 26	R4

This router forwards 20 packets each to 5 hosts. The IP addresses of the hosts are $10.1.1.16, 10.1.1.72, 10.1.1.132, 10.1.1.191,$ and $10.1.1.205$. The number of packets forwarded via the next hop router $\mathrm{R2}$ is __________.

gatecse-2024-set1 numerical-answers computer-networks routing two-marks

Answer key☟

2.28.10 Routing: GATE CSE 2026 | Set 2 | Question: 11

Consider a file of size 4 million bytes being transferred between two hosts connected via a path consisting of three consecutive links of bandwidth 2 Mbps , 500 kbps , and 1 Mbps , respectively. All processing delays and propagation delays are negligible. Assume that there is no other background traffic over the path and no other additional overhead to transfer the file.

Which one of the following is the total time (in seconds) to transfer the file?

Note: $1\mathrm{M} = 10^6, 1\mathrm{k} = 10^3$

A. 731
B. 64
C. 8
D. 16

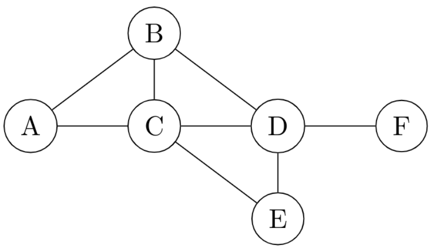
gatecse-2026-set2 computer-networks routing one-mark

Answer key☟

2.28.11 Routing: GATE IT 2005 | Question: 85a

Consider a simple graph with unit edge costs. Each node in the graph represents a router. Each node maintains a routing table indicating the next hop router to be used to relay a packet to its destination and the cost of the path to the destination through that router. Initially, the routing table is empty. The routing table is synchronously updated as follows. In each updated interval, three tasks are performed.

i. A node determines whether its neighbours in the graph are accessible. If so, it sets the tentative cost to each accessible neighbour as 1. Otherwise, the cost is set to ∞.
ii. From each accessible neighbour, it gets the costs to relay to other nodes via that neighbour (as the next hop).
iii. Each node updates its routing table based on the information received in the previous two steps by choosing the minimum cost.



For the graph given above, possible routing tables for various nodes after they have stabilized, are shown in the following options. Identify the correct table.

A. Table for node A

A	-	-
B	B	1
C	C	1
D	B	3
E	C	3
F	C	4

B. Table for node C

A	A	1
B	B	1
C	-	-
D	D	1
E	E	1
F	E	3

C. Table for node B

A	A	1
B	-	-
C	C	1
D	D	1
E	C	2
F	D	2

D. Table for node D

A	B	3
B	B	1
C	C	1
D	-	-
E	E	1
F	F	1

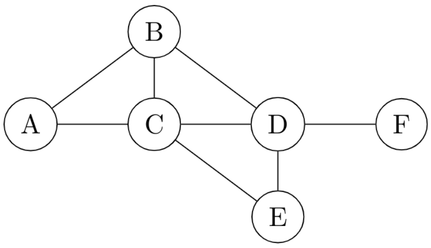
gateit-2005 computer-networks routing normal

Answer key

2.28.12 Routing: GATE IT 2005 | Question: 85b

Consider a simple graph with unit edge costs. Each node in the graph represents a router. Each node maintains a routing table indicating the next hop router to be used to relay a packet to its destination and the cost of the path to the destination through that router. Initially, the routing table is empty. The routing table is synchronously updated as follows. In each updated interval, three tasks are performed.

i. A node determines whether its neighbors in the graph are accessible. If so, it sets the tentative cost to each accessible neighbor as 1. Otherwise, the cost is set to ∞.
ii. From each accessible neighbor, it gets the costs to relay to other nodes via that neighbor (as the next hop).
iii. Each node updates its routing table based on the information received in the previous two steps by choosing the minimum cost.



Continuing from the earlier problem, suppose at some time t, when the costs have stabilized, node A goes down. The cost from node F to node A at time $(t + 100)$ is :

A. > 100 but finite
B. ∞
C. 3
D. > 3 and ≤ 100

gateit-2005 computer-networks routing normal

Answer key

2.28.13 Routing: GATE IT 2007 | Question: 63

A group of 15 routers is interconnected in a centralized complete binary tree with a router at each tree node. Router i communicates with router j by sending a message to the root of the tree. The root then sends the message back down to router j. The mean number of hops per message, assuming all possible router pairs are equally likely is

A. 3
B. 4.26
C. 4.53
D. 5.26

gateit-2007 computer-networks routing binary-tree normal

Answer key

2.28.14 Routing: GATE IT 2008 | Question: 67

Two popular routing algorithms are Distance Vector(DV) and Link State (LS) routing. Which of the following are true?

(S1): Count to infinity is a problem only with DV and not LS routing
(S2): In LS, the shortest path algorithm is run only at one node
(S3): In DV, the shortest path algorithm is run only at one node
(S4): DV requires lesser number of network messages than LS

A. S1, S2 and S4 only
B. S1, S3 and S4 only
C. S2 and S3 only
D. S1 and S4 only

gateit-2008 computer-networks routing normal

Answer key

2.29 Routing Protocols (1)

2.29.1 Routing Protocols: GATE CSE 2025 | Set 2 | Question: 7

Consider the routing protocols given in **List I** and the names given in **List II**:

	List I		List II
(i)	Distance Vector routing	(a)	Bellman-Ford
(ii)	Link state routing	(b)	Dijkstra

For matching of items in **List I** with those in **List II**, which ONE of the following options is CORRECT?

A. (i) - (a) and (ii) - (b)
B. (i) - (a) and (ii) - (a)
C. (i) - (b) and (ii) - (a)
D. (i) - (b) and (ii) - (b)

gatecse2025-set2 computer-networks match-the-following routing-protocols easy one-mark

Answer key

2.30 Sliding Window (16)

✍ **Practice Tests:** Test 1 (15Q) Test 2 (11Q)

2.30.1 Sliding Window: GATE CSE 2003 | Question: 84

Host A is sending data to host B over a full duplex link. A and B are using the sliding window protocol for flow control. The send and receive window sizes are 5 packets each. Data packets (sent only from A to B) are all 1000 bytes long and the transmission time for such a packet is 50 μs. Acknowledgment packets (sent only from B to A) are very small and require negligible transmission time. The propagation delay over the link is 200 μs. What is the maximum achievable throughput in this communication?

A. 7.69×10^6 Bps
B. 11.11×10^6 Bps
C. 12.33×10^6 Bps
D. 15.00×10^6 Bps

gatecse-2003 computer-networks sliding-window normal

Answer key

2.30.2 Sliding Window: GATE CSE 2005 | Question: 25

The maximum window size for data transmission using the selective reject protocol with $n\text{-bit}$ frame sequence numbers is:

A. 2^n
B. 2^{n-1}
C. $2^n - 1$
D. 2^{n-2}

gatecse-2005 computer-networks sliding-window easy

Answer key

2.30.3 Sliding Window: GATE CSE 2006 | Question: 44

Station A uses 32 byte packets to transmit messages to Station B using a sliding window protocol. The round trip delay between A and B is 80 milliseconds and the bottleneck bandwidth on the path between A and B is 128 kbps . What is the optimal window size that A should use?

A. 20
B. 40
C. 160
D. 320

gatecse-2006 computer-networks sliding-window normal

Answer key

2.30.4 Sliding Window: GATE CSE 2006 | Question: 46

Station A needs to send a message consisting of 9 packets to Station B using a sliding window (window size 3) and go-back-n error control strategy. All packets are ready and immediately available for transmission. If every 5th packet that A transmits gets lost (but no acks from B ever get lost), then what is the number of packets that A will transmit for sending the message to B?

A. 12 B. 14 C. 16 D. 18

gatecse-2006 computer-networks sliding-window normal

Answer key

2.30.5 Sliding Window: GATE CSE 2007 | Question: 69

The distance between two stations M and N is L kilometers. All frames are K bits long. The propagation delay per kilometer is t seconds. Let R bits/second be the channel capacity. Assuming that the processing delay is negligible, the minimum number of bits for the sequence number field in a frame for maximum utilization, when the $\text{sliding window protocol}$ is used, is:

A. $\lceil \log_2 \frac{2LtR+2K}{K} \rceil$

B. $\lceil \log_2 \frac{2LtR}{K} \rceil$

C. $\lceil \log_2 \frac{2LtR+K}{K} \rceil$

D. $\lceil \log_2 \frac{2LtR+2K}{2K} \rceil$

gatecse-2007 computer-networks sliding-window normal

Answer key

2.30.6 Sliding Window: GATE CSE 2009 | Question: 57, ISRO2016-75

Frames of 1000 bits are sent over a 10^6 bps duplex link between two hosts. The propagation time is 25 ms. Frames are to be transmitted into this link to maximally pack them in transit (within the link).

What is the minimum number of bits (I) that will be required to represent the sequence numbers distinctly? Assume that no time gap needs to be given between transmission of two frames.

A. $I = 2$ B. $I = 3$ C. $I = 4$ D. $I = 5$

gatecse-2009 computer-networks sliding-window normal isro2016

Answer key

2.30.7 Sliding Window: GATE CSE 2009 | Question: 58

Frames of 1000 bits are sent over a 10^6 bps duplex link between two hosts. The propagation time is $25ms$. Frames are to be transmitted into this link to maximally pack them in transit (within the link).

Let I be the minimum number of bits (I) that will be required to represent the sequence numbers distinctly assuming that no time gap needs to be given between transmission of two frames.

Suppose that the sliding window protocol is used with the sender window size of 2^I, where I is the numbers of bits as mentioned earlier and acknowledgements are always piggy backed. After sending 2^I frames, what is the minimum time the sender will have to wait before starting transmission of the next frame? (Identify the closest choice ignoring the frame processing time)

A. $16ms$ B. $18ms$ C. $20ms$ D. $22ms$

gatecse-2009 computer-networks sliding-window normal

Answer key

2.30.8 Sliding Window: GATE CSE 2014 | Set 1 | Question: 28

Consider a selective repeat sliding window protocol that uses a frame size of 1 KB to send data on a 1.5 Mbps link with a one-way latency of 50 msec. To achieve a link utilization of 60%, the minimum number of bits required to represent the sequence number field is ________.

gatecse-2014-set1 computer-networks sliding-window numerical-answers normal

Answer key

2.30.9 Sliding Window: GATE CSE 2015 | Set 3 | Question: 28

Consider a network connecting two systems located 8000 Km apart. The bandwidth of the network is 500×10^6 bits per second. The propagation speed of the media is 4×10^6 meters per second. It needs to design a Go-Back-N sliding window protocol for this network. The average packet size is 10^7 bits. The network is to be used to its full capacity. Assume that processing delays at nodes are negligible. Then, the minimum size in bits of the sequence number field has to be ______.

gatecse-2015-set3 computer-networks sliding-window normal numerical-answers

Answer key

2.30.10 Sliding Window: GATE CSE 2016 | Set 2 | Question: 55

Consider a 128×10^3 bits/second satellite communication link with one way propagation delay of 150 milliseconds. Selective retransmission (repeat) protocol is used on this link to send data with a frame size of 1 kilobyte. Neglect the transmission time of acknowledgement. The minimum number of bits required for the sequence number field to achieve 100% utilization is ________.

gatecse-2016-set2 computer-networks sliding-window normal numerical-answers

Answer key

2.30.11 Sliding Window: GATE CSE 2026 | Set 1 | Question: 35

Consider the implementation of sliding window protocol over a lossless link, with a window size of W frames, where each frame is of size 1000 bits (including header). The bandwidth of the link is 100kbps $\left(1\text{k} = 10^3\right)$ and the one-way propagation delay is 100 milliseconds. Assume that processing times at the sender and receiver are zero and the transmission time of acknowledgements is also zero. Which one of the following options gives the minimum size of W (in number of frames) required to achieve 100% link utilization?

A. 10
B. 21
C. 20
D. 11

gatecse-2026-set1 two-marks sliding-window computer-networks

Answer key

2.30.12 Sliding Window: GATE IT 2004 | Question: 81

In a sliding window ARQ scheme, the transmitter's window size is N and the receiver's window size is M. The minimum number of distinct sequence numbers required to ensure correct operation of the ARQ scheme is

A. $\min(M, N)$
B. $\max(M, N)$
C. $M + N$
D. MN

gateit-2004 computer-networks sliding-window normal

Answer key

2.30.13 Sliding Window: GATE IT 2004 | Question: 83

A 20 Kbps satellite link has a propagation delay of 400 ms. The transmitter employs the "go back n ARQ" scheme with n set to 10. Assuming that each frame is 100 byte long, what is the maximum data rate possible?

A. 5 Kbps
B. 10 Kbps
C. 15 Kbps
D. 20 Kbps

gateit-2004 computer-networks sliding-window normal

Answer key

2.30.14 Sliding Window: GATE IT 2004 | Question: 88

Suppose that the maximum transmit window size for a TCP connection is 12000 bytes. Each packet consists of 2000 bytes. At some point in time, the connection is in slow-start phase with a current transmit window of 4000 bytes. Subsequently, the transmitter receives two acknowledgments. Assume that no packets are lost and there are no time-outs. What is the maximum possible value of the current transmit window?